

DISAPERE

A Dataset for Discourse Structure in Peer Review Discussions

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University of Massachusetts Amherst

Manuscript

Authors



Peer Discussion Structure in Peer Review Discussions

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Abstract

Any such manuscript is an extensive labeled dataset. While the OpenReview platform (Cheng et al., 2019) has made it easy to build individual public peer review text, labeling this data for use in research requires highly qualified annotations. Correct interpretation of the discussion structure of the text requires an understanding of the technical content, including the use of standard mathematical techniques. Peer work on discourse in peer review has focused more on labeling labels from an labeling arguments extracted from the text, which can be a complex task. If more examples of the structure of research on non-argumentative literature in peer review, then this has been extensive research and deep analysis of different aspects of peer review, the literature needs to be able to understand the discourse relations between reviews and reviews (Cheng et al., 2019; Bar et al., 2019), and some to focus on the structure of the text.

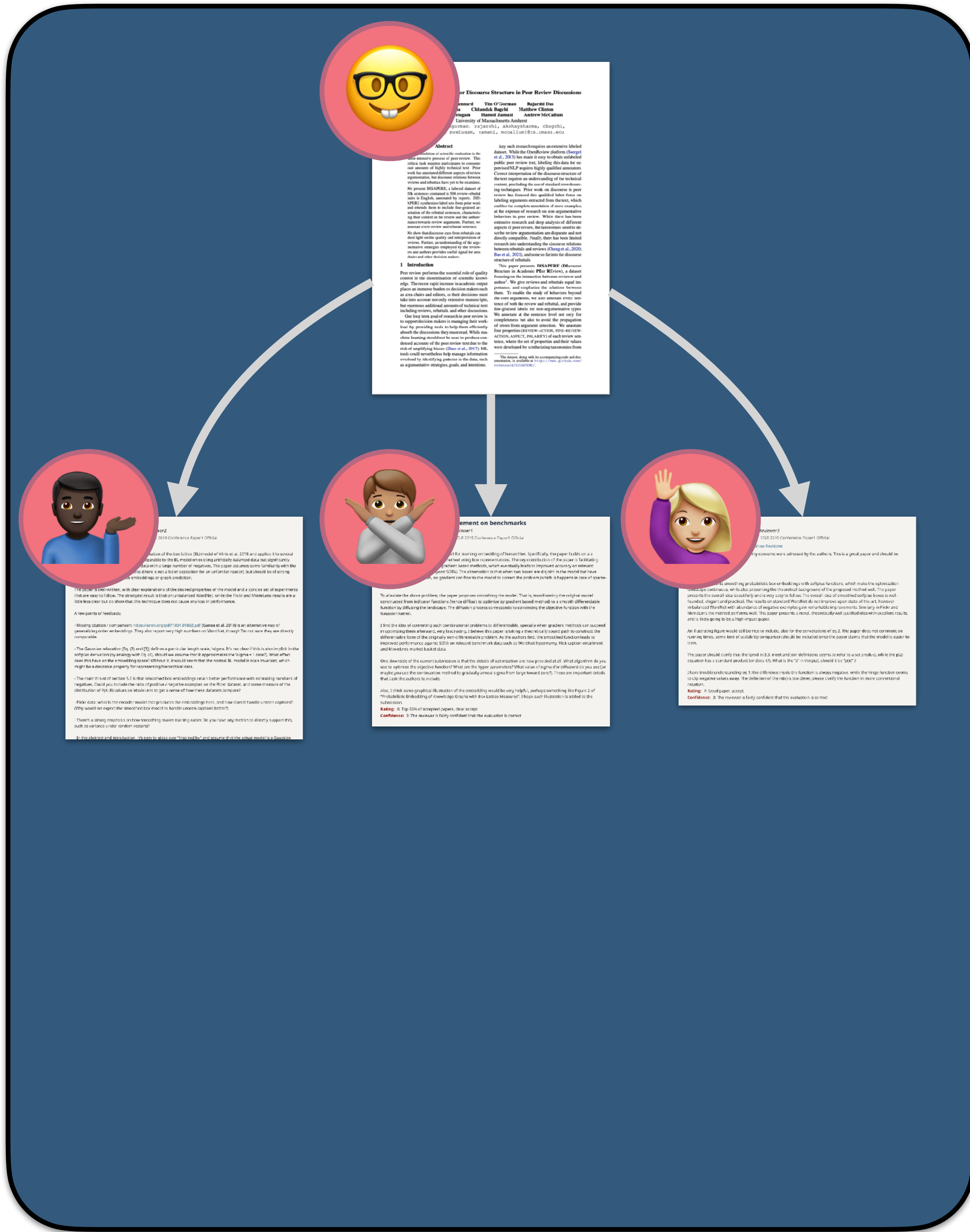
1 Introduction

Peer review performs the essential role of quality control in the dissemination of scientific knowledge. Therefore rapid increase in academic output places an immense burden on decision makers such as area chairs and editors, as their decisions must take into account not only extensive manuscripts, but numerous additional amounts of technical text including reviews, rebuttals, and other discussions. One long term goal of research in peer review is to support decision makers in managing their work load by providing tools to help them efficiently absorb the discussion they generated. While most of the existing research has been in the area of analyzing reviews (Dhar et al., 2017), ML tools could nevertheless help manage information revealed by identifying patterns in the data, such as argumentative structure, goals, and structure.

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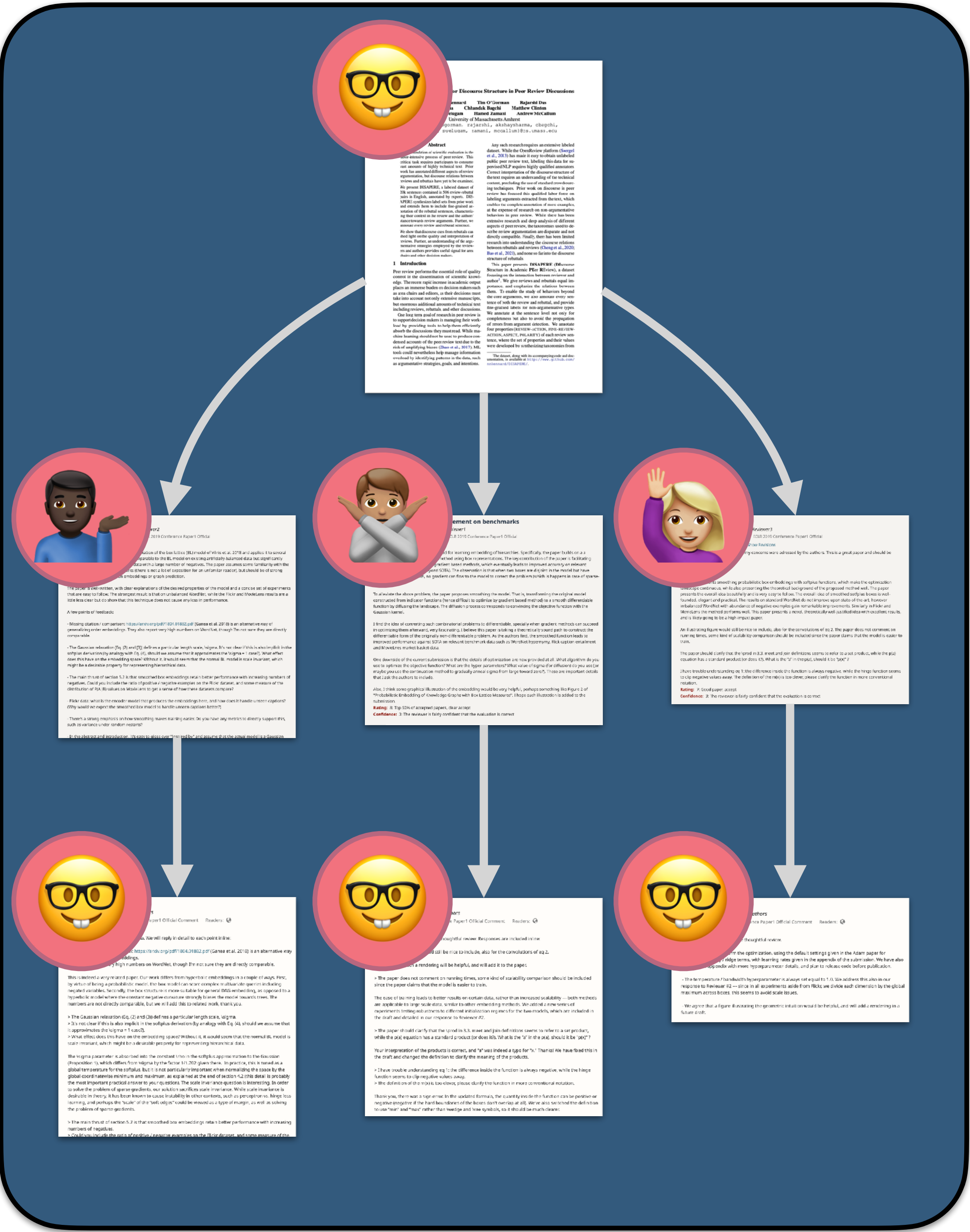
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Reviews

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Rebuttals

Authors



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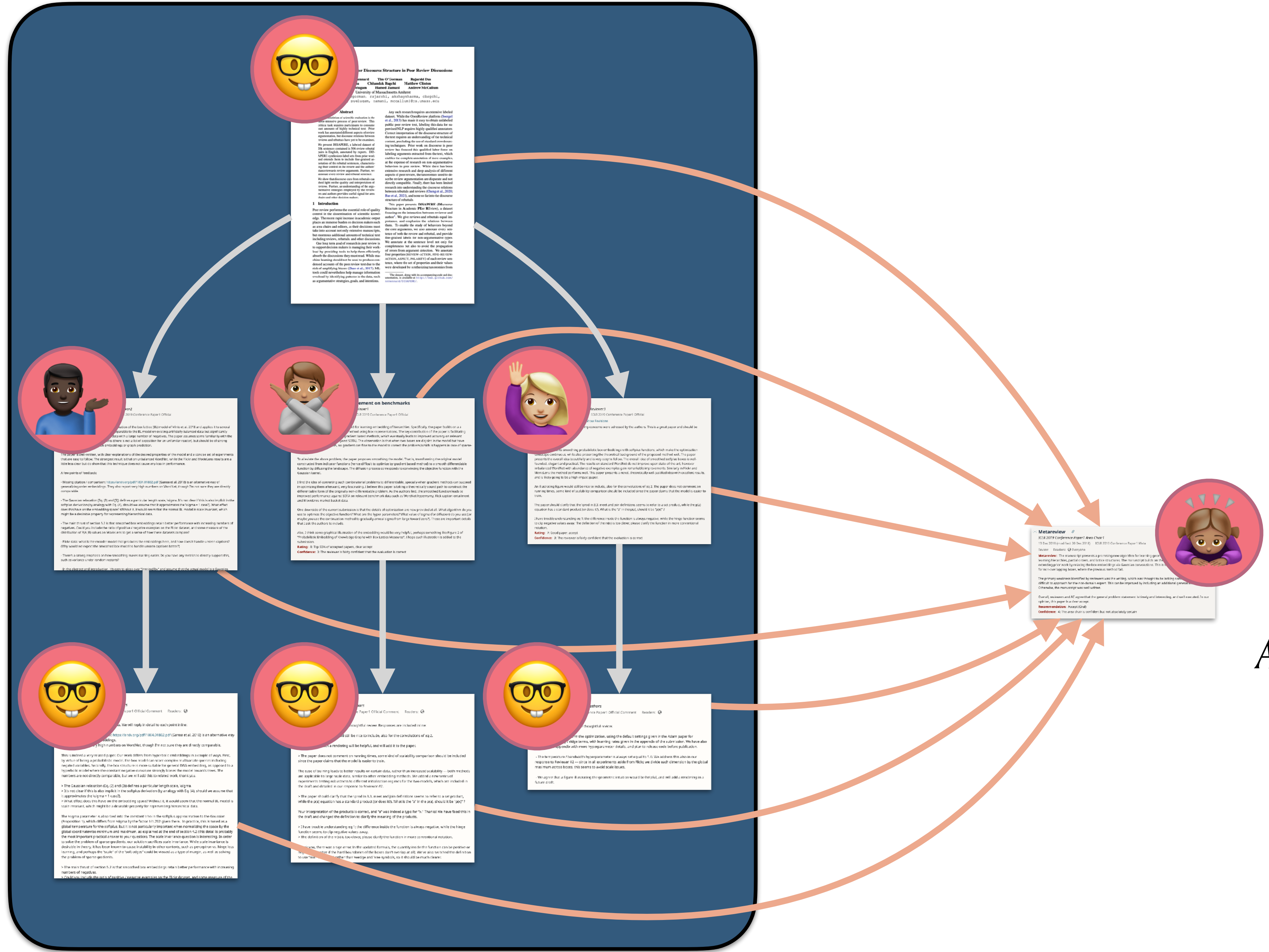
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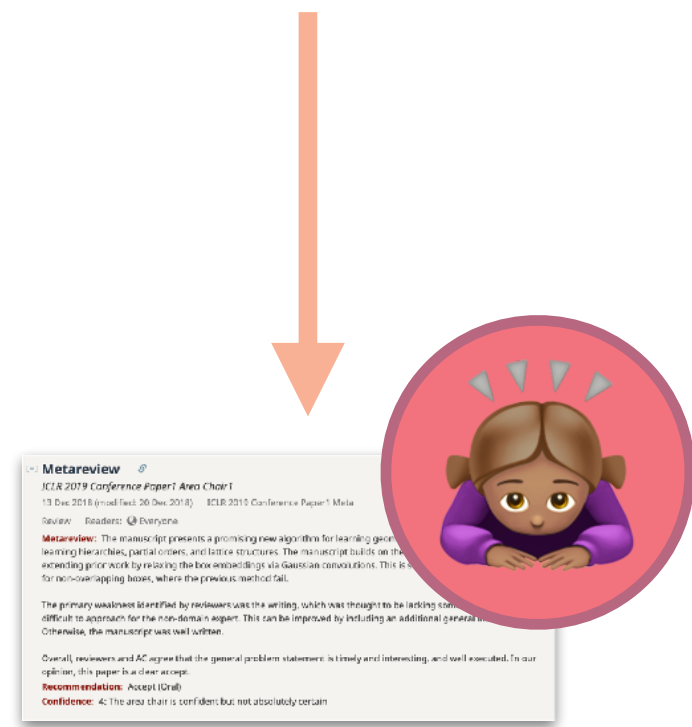
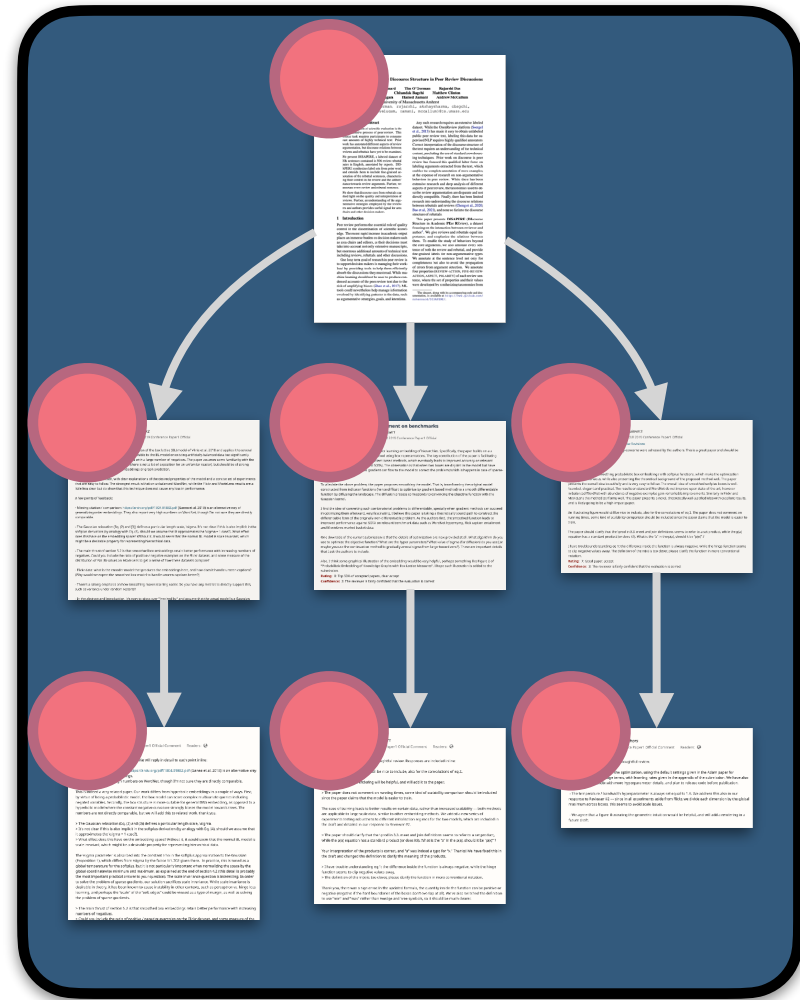
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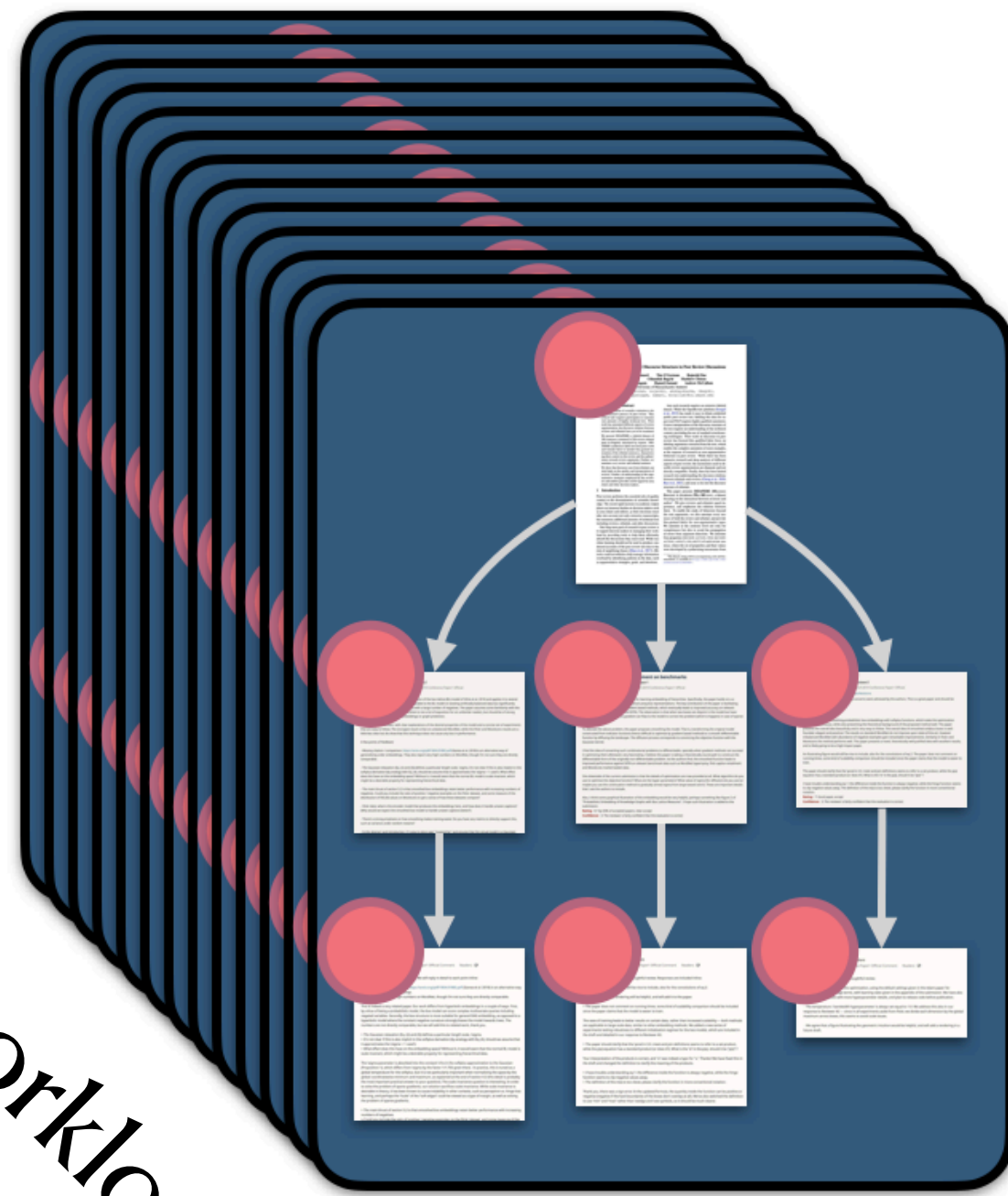
Metareview

Area chair (AC)

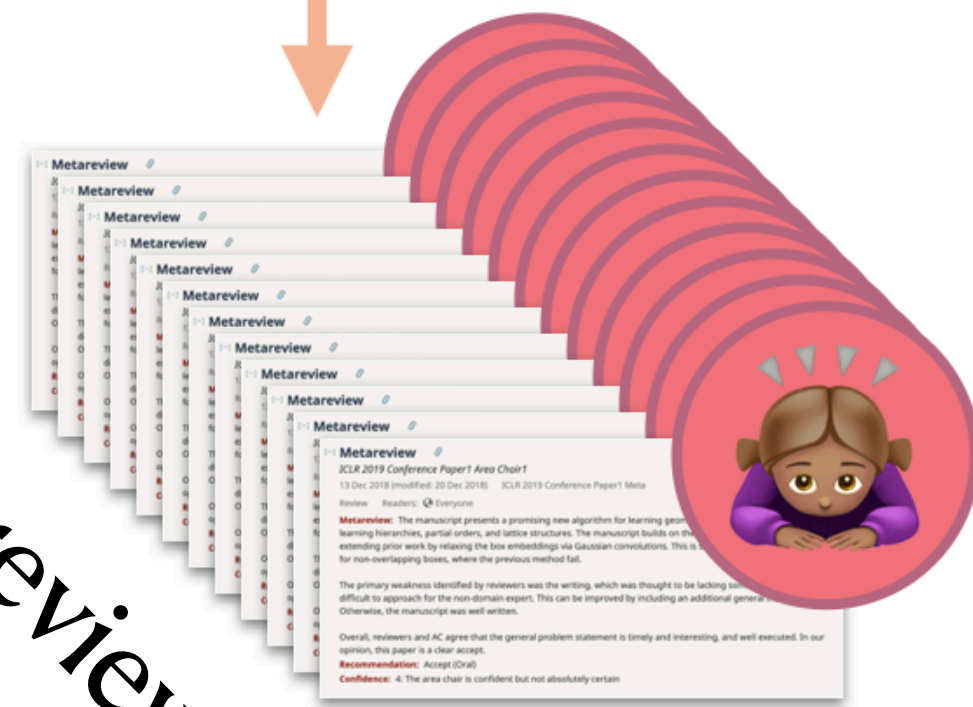




AC workload

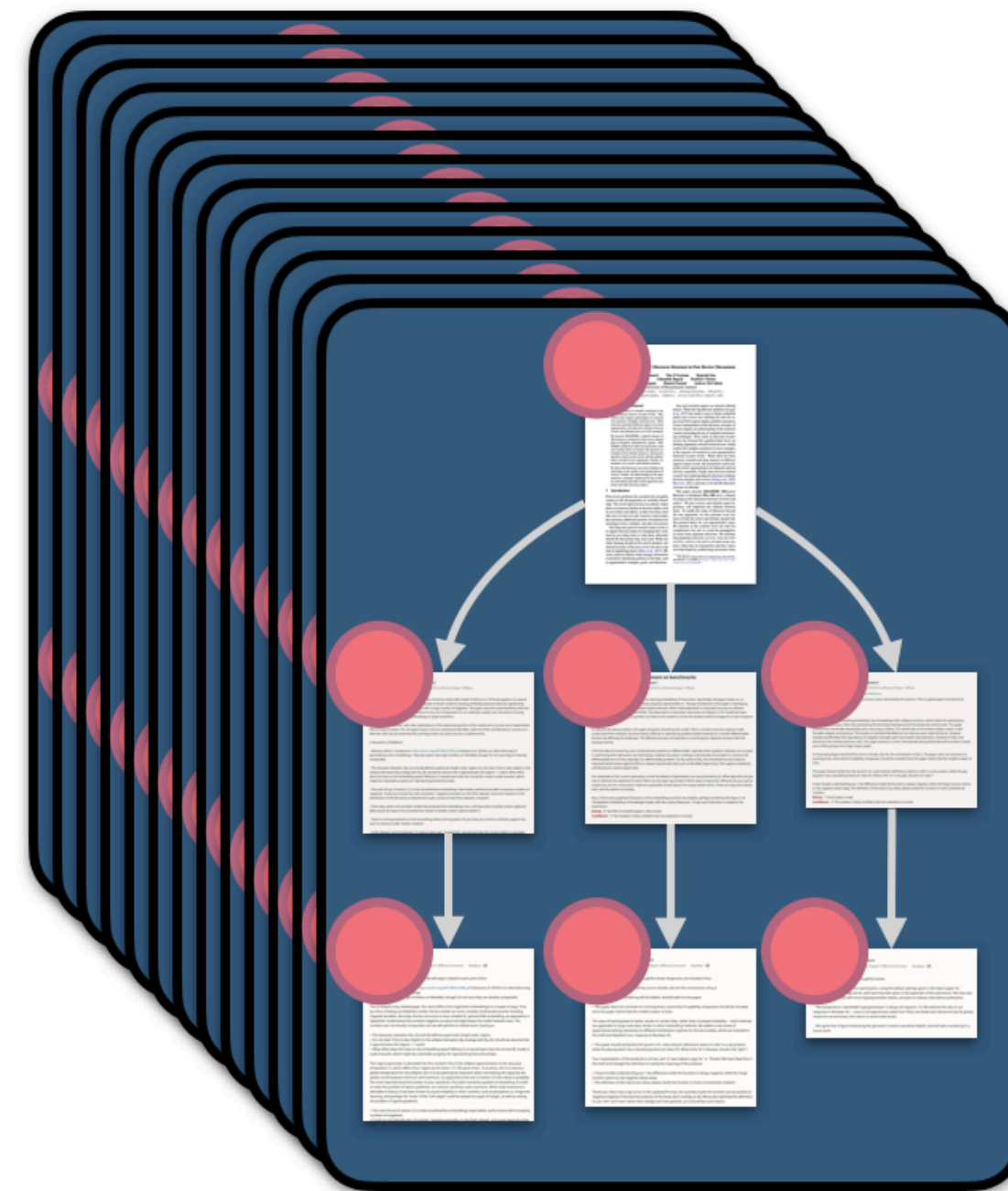
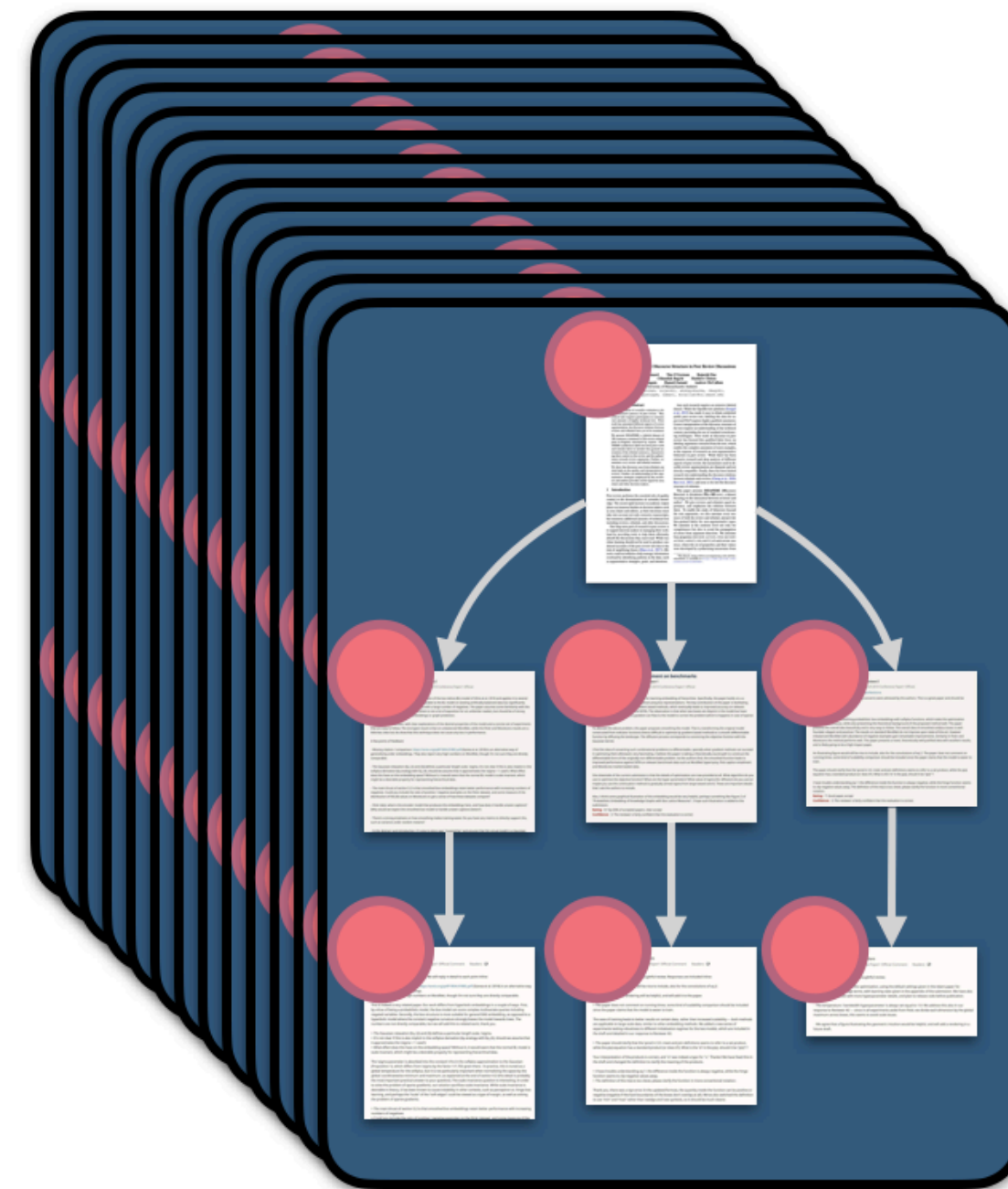
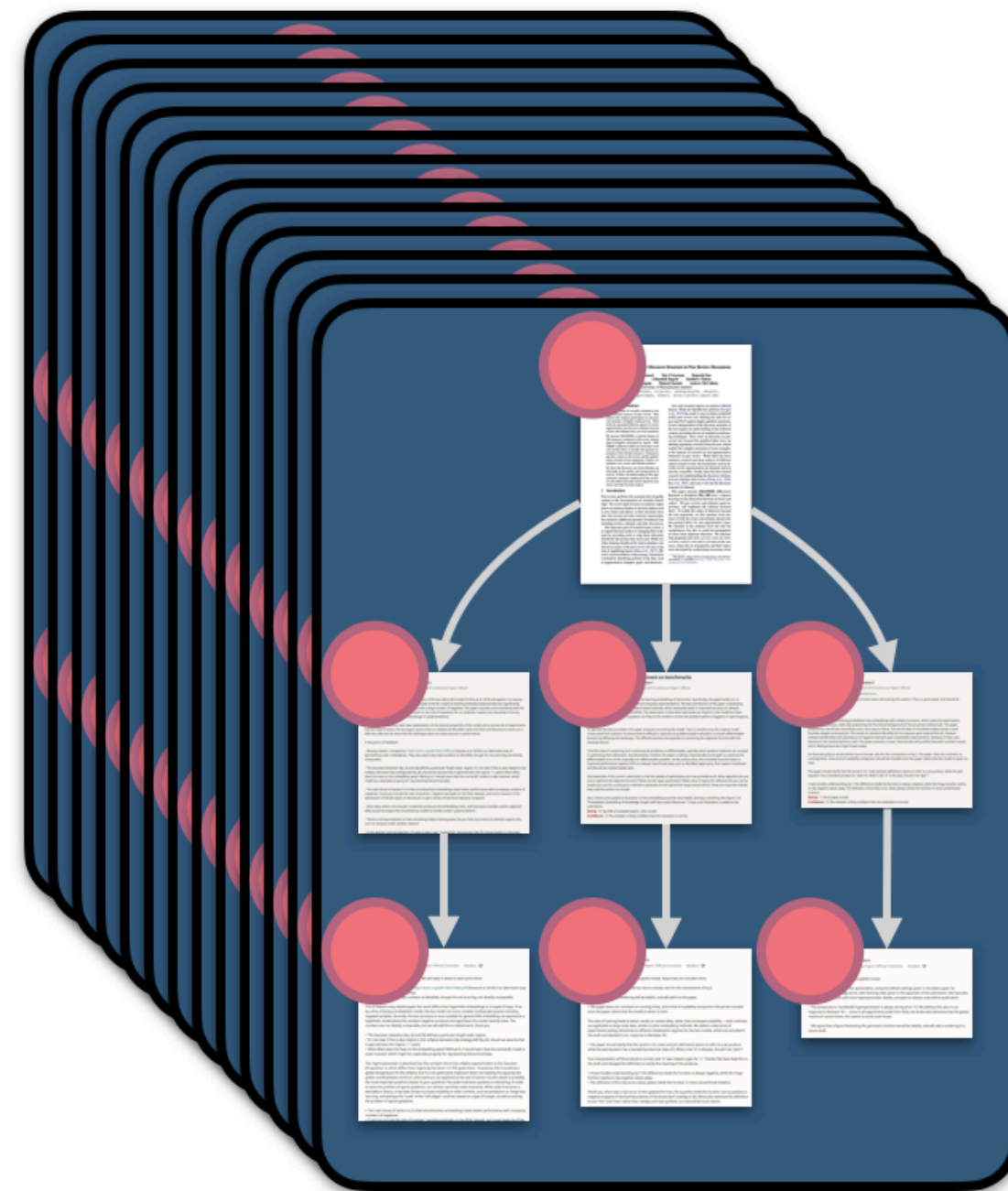
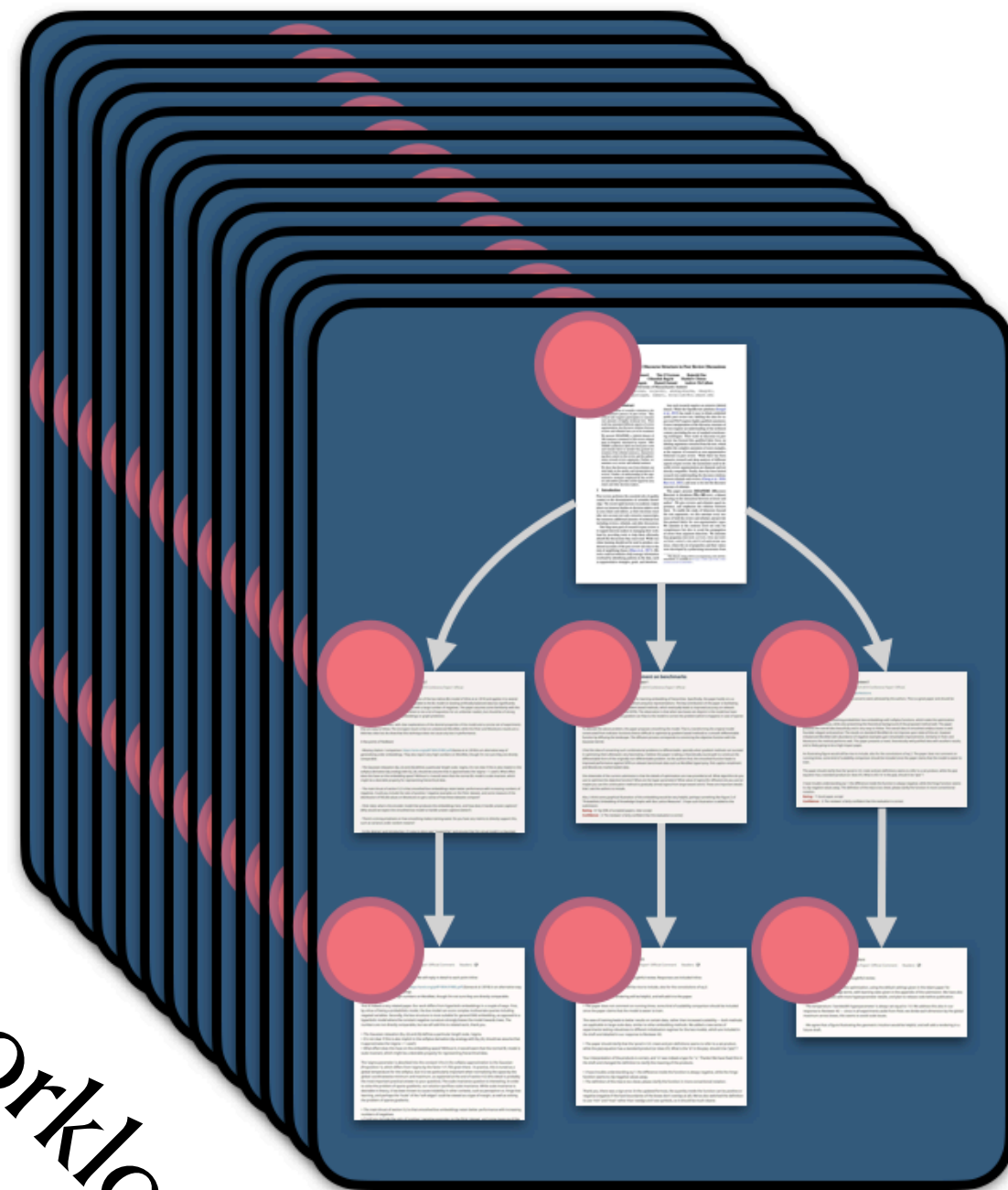


Metareviews

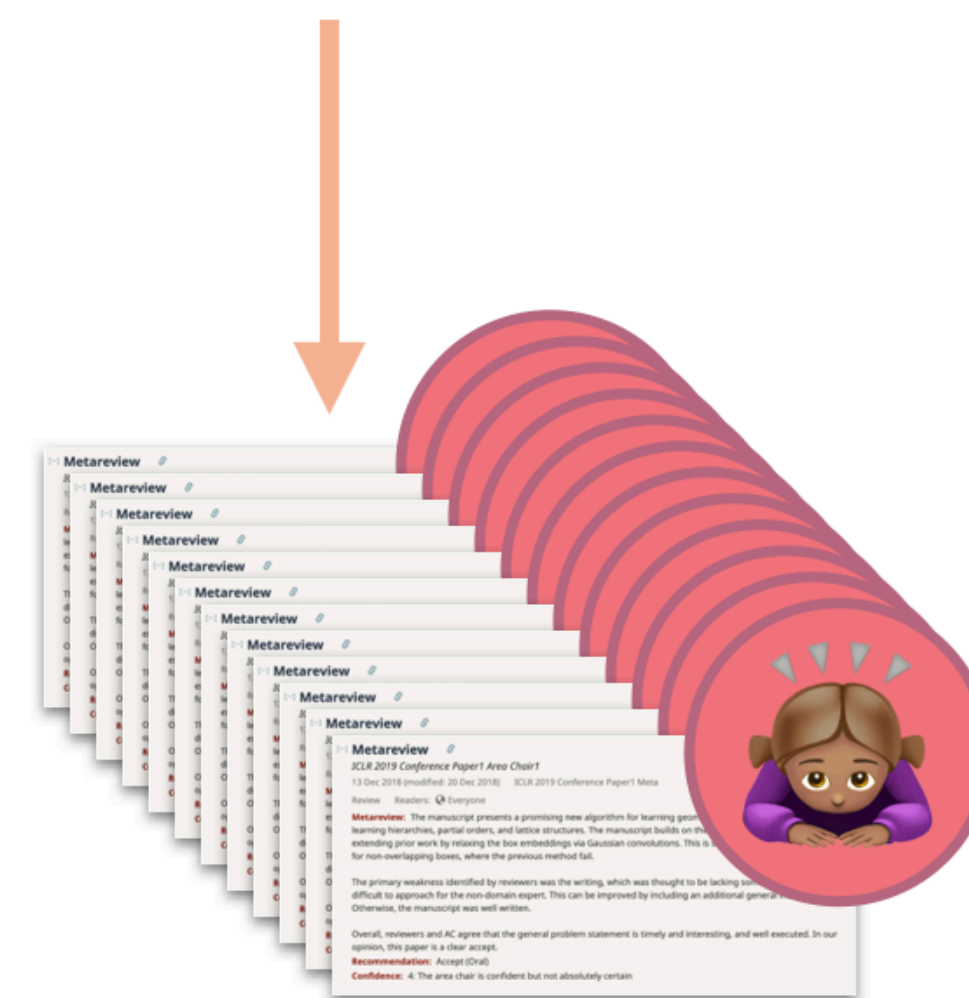
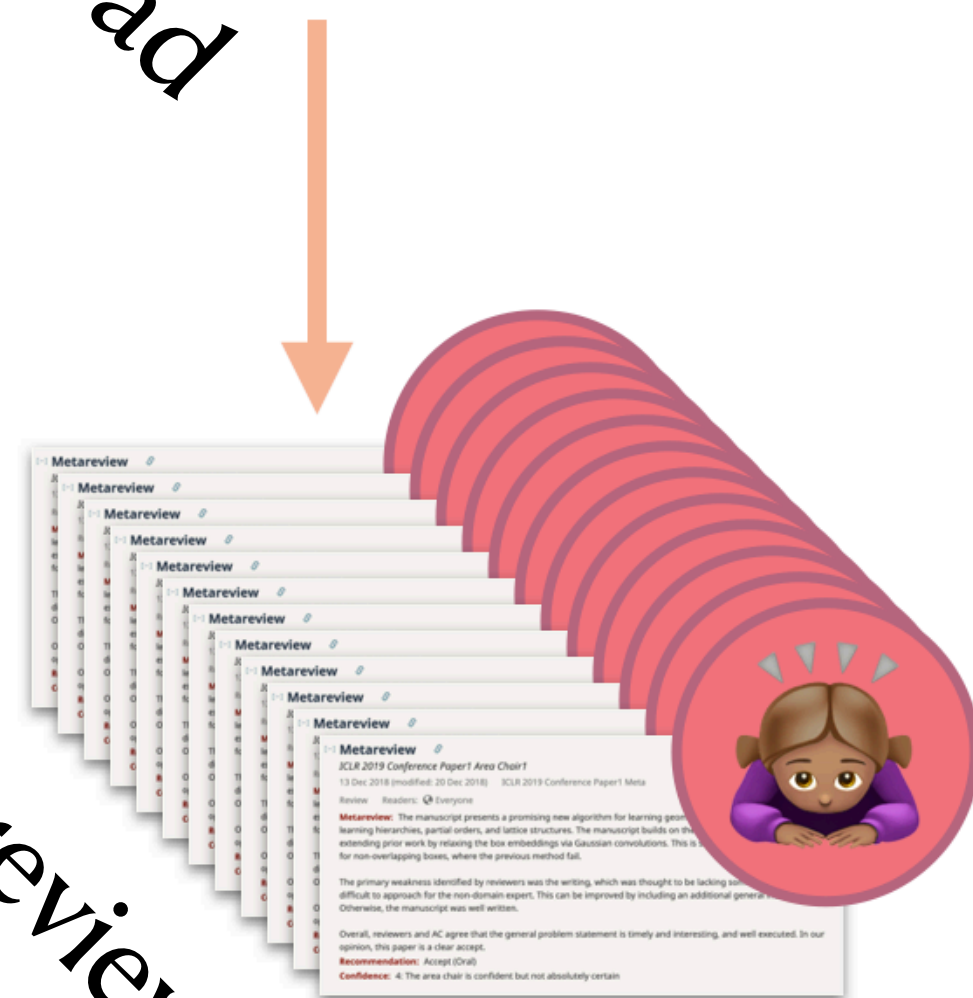


ICLR

AC workload



Metareviews



ICLR

NeurIPS

ARR

ICML

Two important goals of peer review

Two important goals of peer review

Goal
Make correct decisions

Two important goals of peer review

Goal
Make correct decisions
Give constructive feedback

Two important goals of peer review

Goal	Evaluative metric
Make correct decisions	Score variance
Give constructive feedback	

Two important goals of peer review

Goal	Evaluative metric
Make correct decisions	Score variance
Give constructive feedback	?

DISAPERE dataset

Discourse Structure in Academic PEer REview

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DIscourse **S**tructure in **A**cademic **PE**er **RE**view

5 classification tasks, 1 alignment task

506 review-rebuttal pairs

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DIscourse **S**tructure in **A**cademic **PE**er **RE**view

5 classification tasks, 1 alignment task

506 review-rebuttal pairs

Taken from ICLR 2019-2020 ([OpenReview.net](https://openreview.net))

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DIscourse **S**tructure in **A**cademic **PE**er **RE**view

5 classification tasks, 1 alignment task

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Over 21k sentences

Over 850 person-hours of expert annotation

Questions

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Can we measure whether feedback is **constructive**?

Questions

Can we measure whether feedback is **constructive**?

Did the reviewer **try** to be helpful?

Questions

Can we measure whether feedback is **constructive**?

Did the reviewer **try** to be helpful?

Can we measure whether feedback is **applicable**?

Questions

Can we measure whether feedback is **constructive**?

Did the reviewer **try** to be helpful?

Can we measure whether feedback is **applicable**?

Did the reviewer **succeed** in being helpful?

Is the feedback constructive?

Sentence-level review labels

Argument Mining for Understanding Peer Reviews (Hua et al., NAACL 2019)

Can We Automate Scientific Reviewing? (Yuan et al., arXiv 2021)

Argument Mining Driven Analysis of Peer-Reviews (Fromm et al., arXiv 2021)

Sentence-level review labels

Argumentative

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Sentence-level review labels

Evaluative

It's hard to evaluate whether the method would be useful.

Argumentative

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Sentence-level review labels

<i>Argumentative</i>	Evaluative	<i>It's hard to evaluate whether the method would be useful.</i>
	Request	<i>How does the number of layers affect performance?</i>

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Sentence-level review labels

Argumentative

Evaluative

It's hard to evaluate whether the method would be useful.

Request

How does the number of layers affect performance?

Fact

This is becoming a crowded space, with many such models.

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24 Oct 2019 (modified: 05 Nov 2019)ICLR 2020 Conference Paper443 Official Review

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24 Oct 2019 (modified: 05 Nov 2019) ICLR 2020 Conference Paper443 Official Review Readers: Everyone

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I believe that the presentation of the proposed method can be significantly improved. The method description was a bit confusing and unclear to me. The experimental results presented were all done on small synthetic datasets and it's hard to evaluate whether the method is practically useful. Furthermore, no comparisons were provided to any baselines/alternative methods. For example, in Sections 4 and 5 I was hoping to see comparisons to methods like MAML. Also, I felt that the proposed approach in Section 5 is very similar to MAML intuitively. This makes a comparison with MAML even more desirable. Without any comparisons it's hard to tell how difficult the tasks under consideration are and what would amount to good performance on the held-out tasks.

In summary, I feel the paper tackles an interesting problem with an interesting approach, but the content could be organized much better. Also, this work would benefit significantly from a better experimental evaluation. For these reasons I lean towards rejecting this paper for now, but would love to see it refined for a future machine learning conference.

Also, the work by Platanios, et al. on contextual parameter generation is very relevant to this work as it tackles multi-task learning using HyperNetworks. It may be worth adding a short discussion/comparison to that work as it also considers zero-shot learning.

Minor comments:
- Capitalize: "section" -> "Section", "appendix" -> "Appendix", "fig." -> "Figure". Sometimes these are capitalized, but the use is inconsistent throughout the paper.
- "Hold-out" vs "held-out". Be consistent and use "held-out" throughout.

Rating: 3: Weak Reject
Experience Assessment: I have published one or two papers in this area.
Review Assessment: Thoroughness In Paper Reading: N/A
Review Assessment: Checking Correctness Of Derivations And Theory: N/A
Review Assessment: Checking Correctness Of Experiments: I assessed the sensibility of the experiments.

Reviewer recommendation:
3 (Reject)

“I hope that an updated
version will be accepted
elsewhere.”

8/21 sentences are requests

This paper presents...	Structuring	
	Structuring	
The premise is very interesting...	Evaluative	
	Request	←
	Evaluative	
... whether the model is practically useful...	Evaluative	
	Evaluative	
I was hoping to see...	Request	←
	Request	←
... it’s hard to tell how difficult the tasks...	Evaluative	
	Evaluative	
... work would benefit from better evaluation...	Request	←
	Social	
	Fact	
... it may be worth adding a discussion of...	Request	←
	Structuring	
	Request	←
	Evaluative	
... capitalization...	Request	←
Be consistent about...	Request	←

[-]Official Blind Review #2

ICLR 2020 Conference Paper1166 AnonReviewer2

23 Oct 2019 (modified: 05 Nov 2019)ICLR 2020 Conference Paper1166 Official Review

Readers:  Everyone

Review: Summary:

This paper describes a contextual encoding scheme for reconstruction of 3D pointclouds from 2D images. An encoder outputs the parameters of a hierarchy of reconstruction networks that can be applied in succession to map random samples on a unit sphere to the surface of the reconstructed shape.

Strengths:

The author's model was quite novel in my opinion. Deep 2D->3D is becoming a crowded space and there are many other models that encode image inputs, and many others that perform recursive or composition-based decoding. However, the particular link here was interesting, and I appreciate the small number of parameters resulting in solid reconstruction performance. While most related work was covered well, I believe the authors could have a more up-to-date list of recent work that reconstructs triangle-mesh representations from images [A-C] (especially since several of these methods has an architecture that involves encoding and subsequent compositional refinement).

Some of the reconstructions shown in this paper are quite impressive, and the quantitative results show outperforming 2 recent methods. I did appreciate also the novel path-based evaluation of shape accuracy in the Appendix, although it would have been helpful to see more discussion of this in the main paper.

Areas for improvement:

I found that the core technical description was quite brief and would have benefited from simply more detail and space. You have argued that your method is sensible to try (cog. sci motivations), and shown that one instance works, but what can we expect in a more mathematical or general sense? Can any sizes of encoder and mapping network fit together? How does the number of mapping layers effect performance? Won't we eventually expect vanishing/exploding gradients with particular activation and can one address this in some way?

I note that recent papers in this field tend to perform significantly more extensive experimental evaluation, typically selecting a wider range of competitors and using a number of more standardized metrics including IOU, F1 score and CD and typically repeating these at a variety of resolutions or on additional datasets or category splits etc.

Decision:

Weak reject because the idea is quite interesting, but I believe a more thorough explanation and expanded experimental comparison would be of great help to ensure the community can appreciate this work.

Additional citations suggested:

[A] Pixel2Mesh: Generating 3D Mesh Models from Single RGB Images. Wang, Zhang, Li, Fu, Liu and Jiang. ECCV 2018.
[B] MeshCNN: A Network with an Edge. Hanocka, Hertz, Fish, Giryas, Fleishman and Cohen-Or. SIGGRAPH 2019.
[C] GEOMetrics: Exploiting Structure for Graph-Encoded Objects. Smith, Fujimoto, Romero and Meger. ICML 2019.

Rating: 3: Weak Reject

Experience Assessment:

I have published in this field for several years.

Review Assessment: Thoroughness In Paper Reading:

I read the paper thoroughly.

Review Assessment: Checking Correctness Of Derivations And Theory:

I carefully checked the derivations and theory.

Review Assessment: Checking Correctness Of Experiments:

I carefully checked the experiments.

Reviewer recommendation:
3 (Reject)

[-]Official Blind Review #2

ICLR 2020 Conference Paper1166 AnonReviewer2

23 Oct 2019 (modified: 05 Nov 2019)ICLR 2020 Conference Paper1166 Official ReviewReaders:  Everyone

Review:

Summary:

This paper describes a contextual encoding scheme for reconstruction of 3D pointclouds from 2D images. An encoder outputs the parameters of a hierarchy of reconstruction networks that can be applied in succession to map random samples on a unit sphere to the surface of the reconstructed shape.

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[C] GEOMetrics: Exploiting Structure for Graph-Encoded Objects. Smith, Fujimoto, Romero and Meger. ICML 2019.

Rating:

3: Weak Reject

Experience Assessment:

I have published in this field for several years.

Review Assessment: Thoroughness In Paper Reading:

I read the paper thoroughly.

Review Assessment: Checking Correctness Of Derivations And Theory:

I carefully checked the derivations and theory.

Review Assessment: Checking Correctness Of Experiments:

I carefully checked the experiments.

Reviewer recommendation:
3 (Reject)

7/20 sentences are requests

Structuring
Structuring
Structuring
Structuring
Evaluative
Fact
Evaluative
Request
Evaluative
Evaluative
Structuring
Request
Request
Fact
Structuring
Request
Structuring
Request
Request
Request

Official Blind Review #2

ICLR 2020 Conference Paper1166 AnonReviewer2

23 Oct 2019 (modified: 05 Nov 2019)ICLR 2020 Conference Paper1166 Official ReviewReaders: Everyone

Review:

Summary:

This paper describes a contextual encoding scheme for reconstruction of 3D pointclouds from 2D images. An encoder outputs the parameters of a hierarchy of reconstruction networks that can be applied in succession to map random samples on a unit sphere to the surface of the reconstructed shape.

Strengths:

The author's model was quite novel in my opinion. Deep 2D->3D is becoming a crowded space and there are many other models that encode image inputs, and many others that perform recursive or composition-based decoding. However, the particular link here was interesting, and I appreciate the small number of parameters resulting in solid reconstruction performance. While most related work was covered well, I believe the authors could have a more up-to-date list of recent work that reconstructs triangle-mesh representations from images [A-C] (especially since several of these methods has an architecture that involves encoding and subsequent compositional refinement).

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I note that recent papers in this field tend to perform significantly more extensive experimental evaluation, typically selecting a wider range of competitors and using a number of more standardized metrics including IOU, F1 score and CD and typically repeating these at a variety of resolutions or on additional datasets or category splits etc.

Decision:

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[C] GEOMetrics: Exploiting Structure for Graph-Encoded Objects. Smith, Fujimoto, Romero and Meger. ICML 2019.

Rating:

3: Weak Reject

Experience Assessment:

I have published in this field for several years.

Review Assessment: Thoroughness In Paper Reading:

I read the paper thoroughly.

Review Assessment: Checking Correctness Of Derivations And Theory:

I carefully checked the derivations and theory.

Review Assessment: Checking Correctness Of Experiments:

I carefully checked the experiments.

Reviewer recommendation:
3 (Reject)

7/20 sentences are requests

This paper describes...	Structuring
...	Structuring
...	Structuring
...	Structuring
... was quite novel ...	Evaluative
...	Fact
...	Evaluative
... could highlight more recent work...	Request
...	Evaluative
I did appreciate...	Evaluative
...	Structuring
... benefit from more detail...	Request
...	Request
...	Fact
...	Structuring
...	Request
Additional citations suggested:	Structuring
[1]	Request
[2]	Request
[3]	Request

Official Blind Review #2

ICLR 2020 Conference Paper1166 AnonReviewer2

23 Oct 2019 (modified: 05 Nov 2019) ICLR 2020 Conference Paper1166 Official Review Readers: Everyone

Review: Summary:
This paper describes a contextual encoding scheme for reconstruction of 3D pointclouds from 2D images. An encoder outputs the parameters of a hierarchy of reconstruction networks that can be applied in succession to map random samples on a unit sphere to the surface of the reconstructed shape.

Strengths:
The author's model was quite novel in my opinion. Deep 2D->3D is becoming a crowded space and there are many other models that encode image inputs, and many others that perform recursive or composition-based decoding. However, the particular link here was interesting, and I appreciate the small number of parameters resulting in solid reconstruction performance. While most related work was covered well, I believe the authors could have a more up-to-date list of recent work that reconstructs triangle-mesh representations from images [A-C] (especially since several of these methods has an architecture that involves encoding and subsequent compositional refinement).

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I note that recent papers in this field tend to perform significantly more extensive experimental evaluation, typically selecting a wider range of competitors and using a number of more standardized metrics including IOU, F1 score and CD and typically repeating these at a variety of resolutions or on additional datasets or category splits etc.

Decision:
Weak reject because the idea is quite interesting, but I believe a more thorough explanation and expanded experimental comparison would be of great help to ensure the community can appreciate this work.

Additional citations suggested:

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[C] GEOMetrics: Exploiting Structure for Graph-Encoded Objects. Smith, Fujimoto, Romero and Meger. ICML 2019.

Rating: 3: Weak Reject

Experience Assessment: I have published in this field for several years.

Review Assessment: Thoroughness In Paper Reading: I read the paper thoroughly.

Review Assessment: Checking Correctness Of Derivations And Theory: I carefully checked the derivations and theory.

Review Assessment: Checking Correctness Of Experiments: I carefully checked the experiments.

Reviewer recommendation:
3 (Reject)

7/20 sentences are requests

This paper describes...	Structuring
...	Structuring
...	Structuring
...	Structuring
... was quite novel ...	Evaluative
...	Fact
...	Evaluative
... could highlight more recent work...	Request
...	Evaluative
I did appreciate...	Evaluative
...	Structuring
... benefit from more detail...	Request
...	Request
...	Fact
...	Structuring
...	Request
Additional citations suggested:	Structuring
[1]	Request
[2]	Request
[3]	Request

Fine-grained request labels

Evaluative

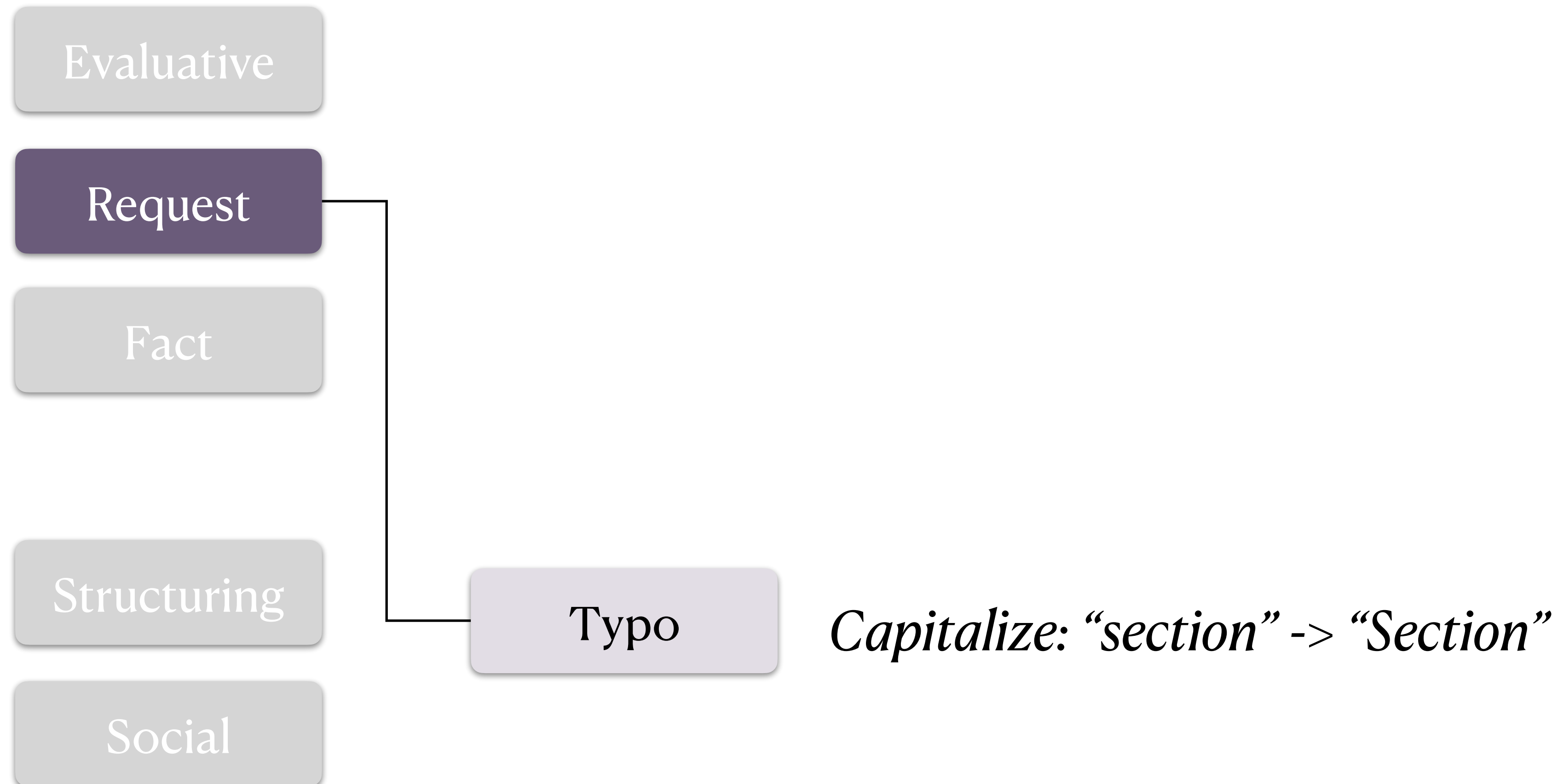
Request

Fact

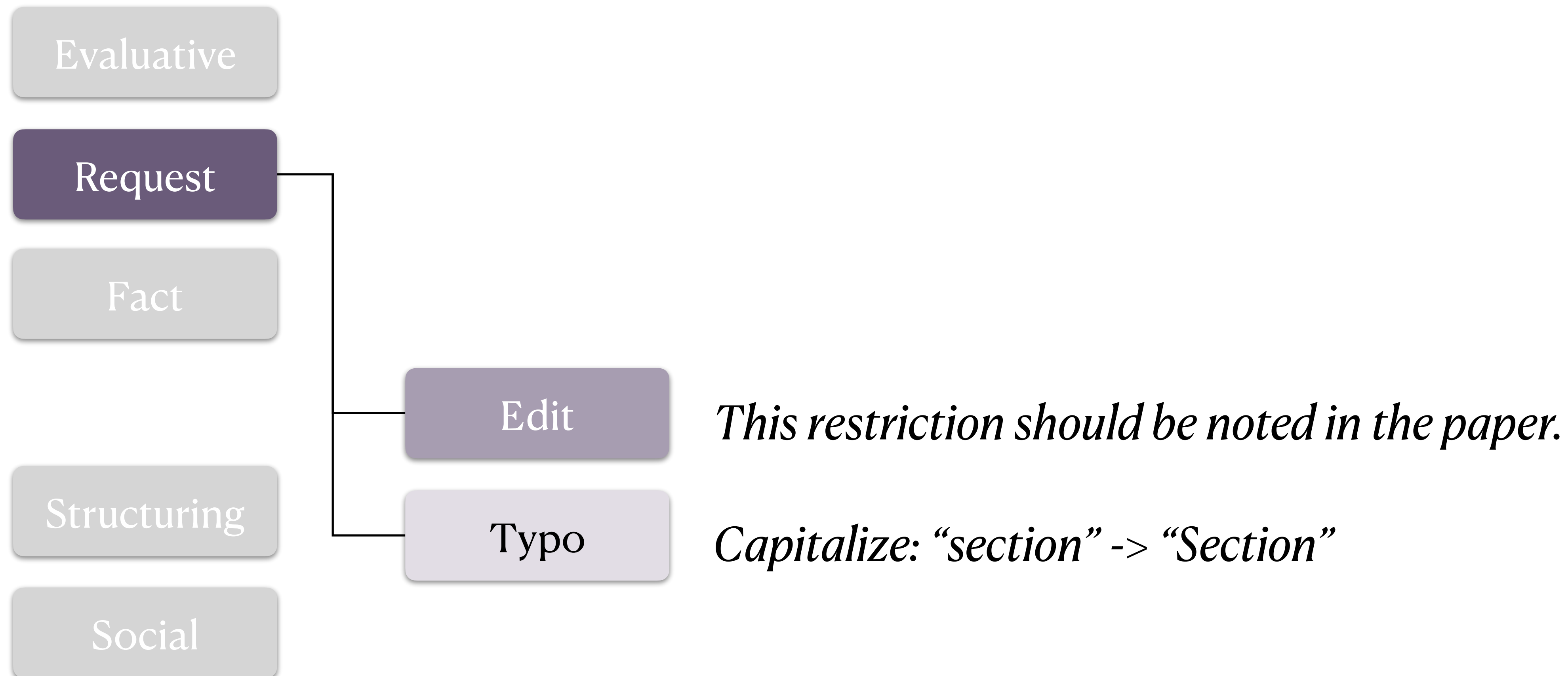
Structuring

Social

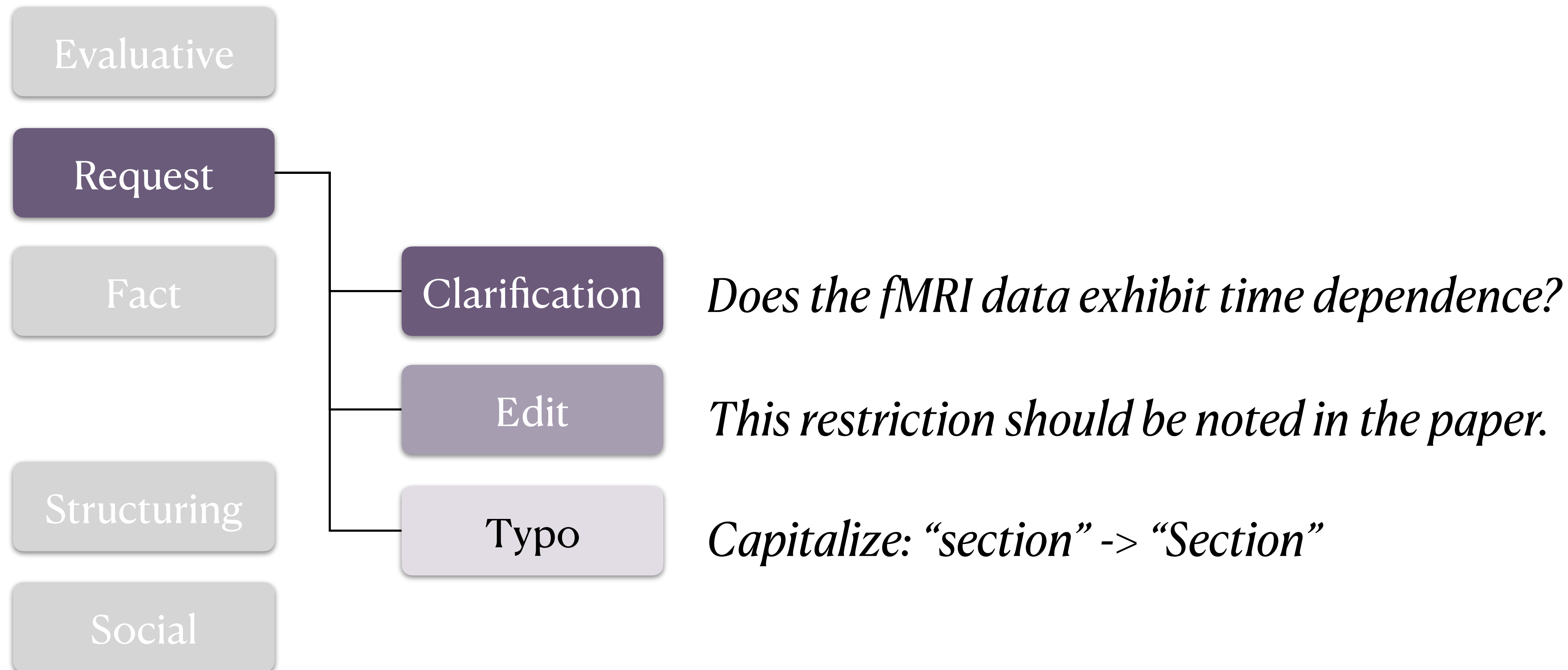
Fine-grained request labels



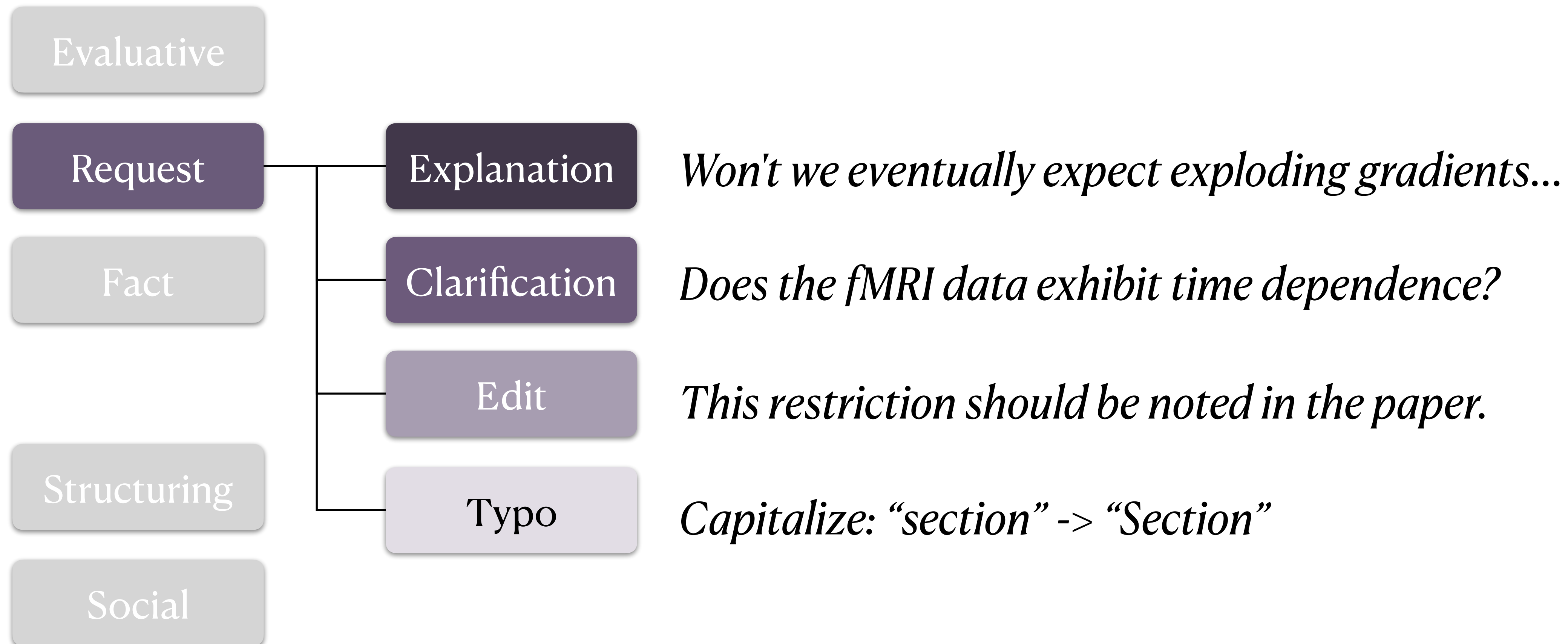
Fine-grained request labels



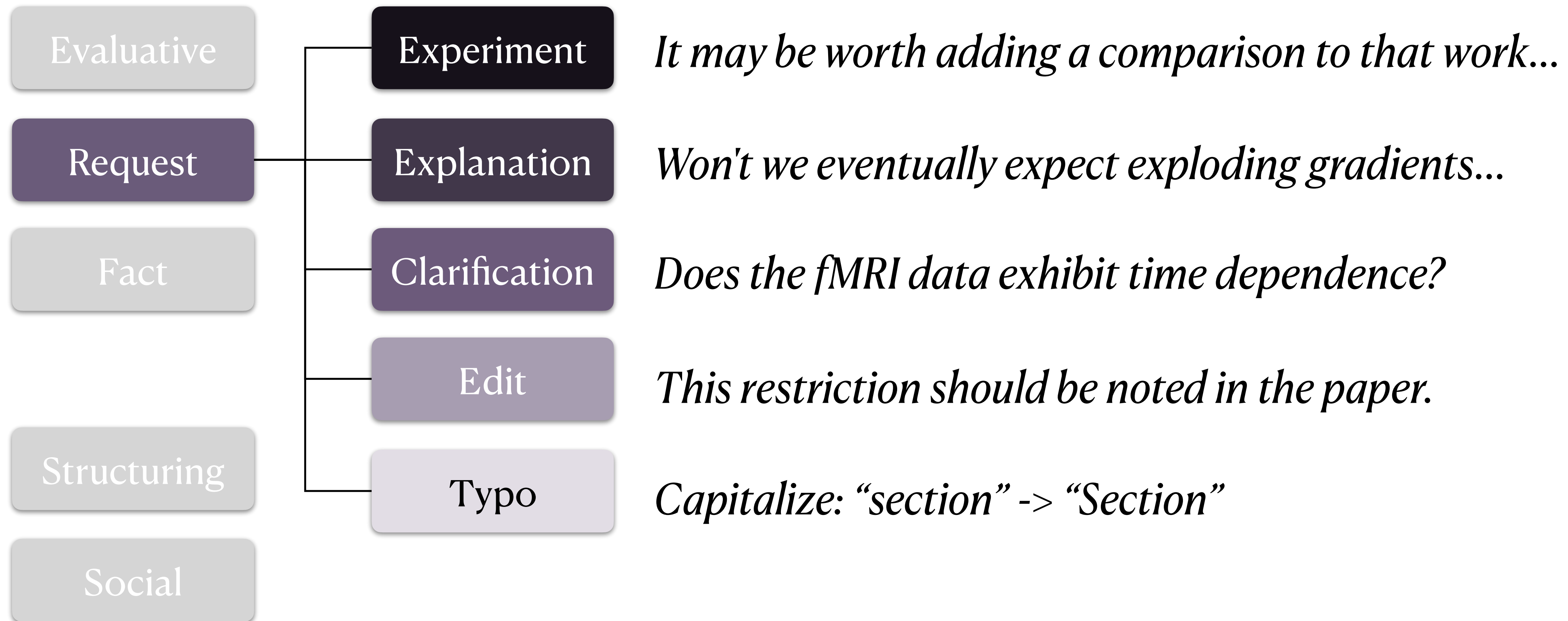
Fine-grained request labels



Fine-grained request labels



Fine-grained request labels

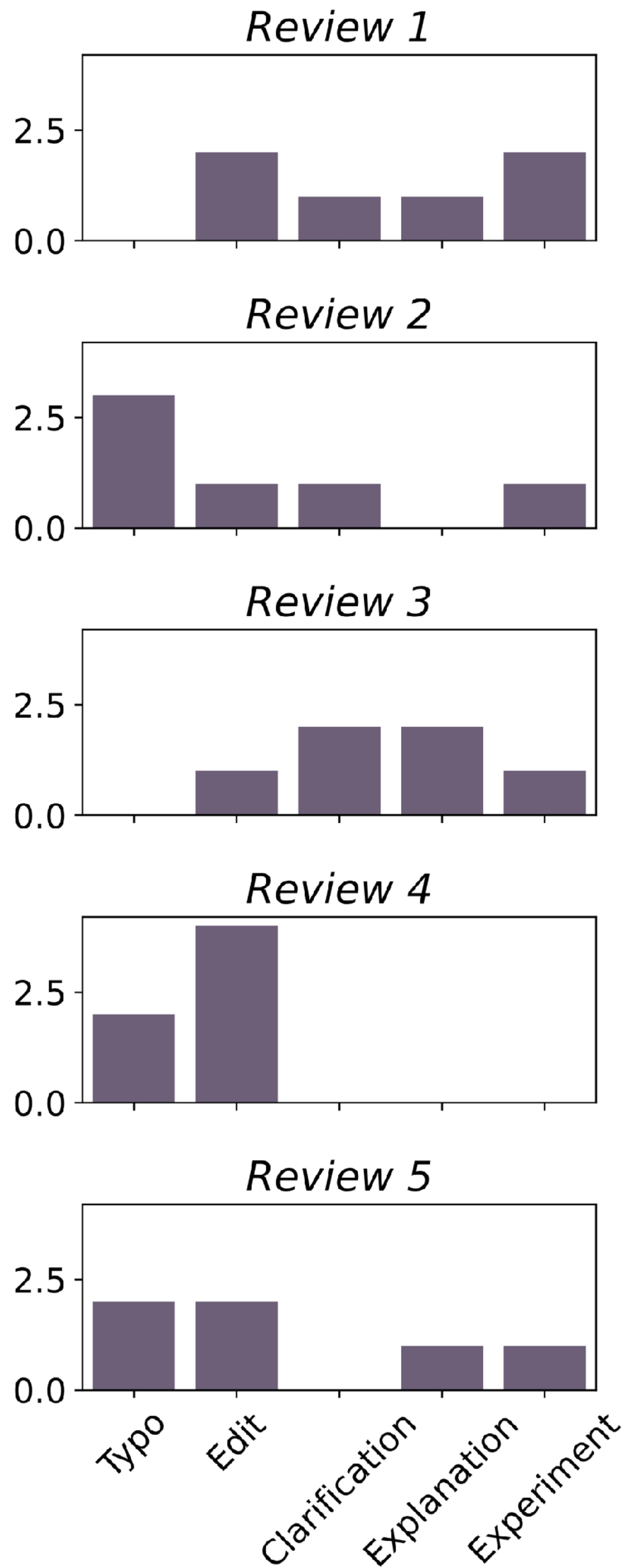


Reviewer recommendation:
3 (Reject)

*6 request statements per
review*

*Reviewer recommendation:
3 (Reject)*

*6 request statements per
review*



What about:

What about:

Questions based on a misunderstanding?

What about:

Questions based on a misunderstanding?

Requests for experiments that are just out of scope?

What about:

Questions based on a misunderstanding?

Requests for experiments that are just out of scope?

Idea: Use the rebuttal, which contains
authors' opinion on the review

Sentence-level rebuttal labels

Sentence-level rebuttal labels

Alignment: context sentences in the review

Sentence-level rebuttal labels

Alignment: context sentences in the review

16 response types

Sentence-level rebuttal labels

Alignment: context sentences in the review

16 response types

Two categories: *concur* and *dispute*

Rebuttal stance: criticizing the criticism

Rebuttal stance: criticizing the criticism

Request

Add comparison to the following baselines...

Rebuttal stance: criticizing the criticism

Request

Add comparison to the following baselines...

Concur

Will be done by camera ready deadline

Rebuttal stance: criticizing the criticism

Request

Add comparison to the
following baselines...

Concur

Will be done by camera ready deadline

Dispute

Out of scope

Rebuttal stance: criticizing the criticism

Request

Add comparison to the following baselines...

Concur

Will be done by camera ready deadline

Dispute

Out of scope

Evaluative

This method might not be practically useful, since...

Rebuttal stance: criticizing the criticism

Request

Add comparison to the following baselines...

Concur

Will be done by camera ready deadline

Dispute

Out of scope

Evaluative

This method might not be practically useful, since...

Concur

Concede criticism

Rebuttal stance: criticizing the criticism

Request

Add comparison to the following baselines...

Concur

Will be done by camera ready deadline

Dispute

Out of scope

Evaluative

This method might not be practically useful, since...

Concur

Concede criticism

Dispute

Mitigate criticism

Agreeability

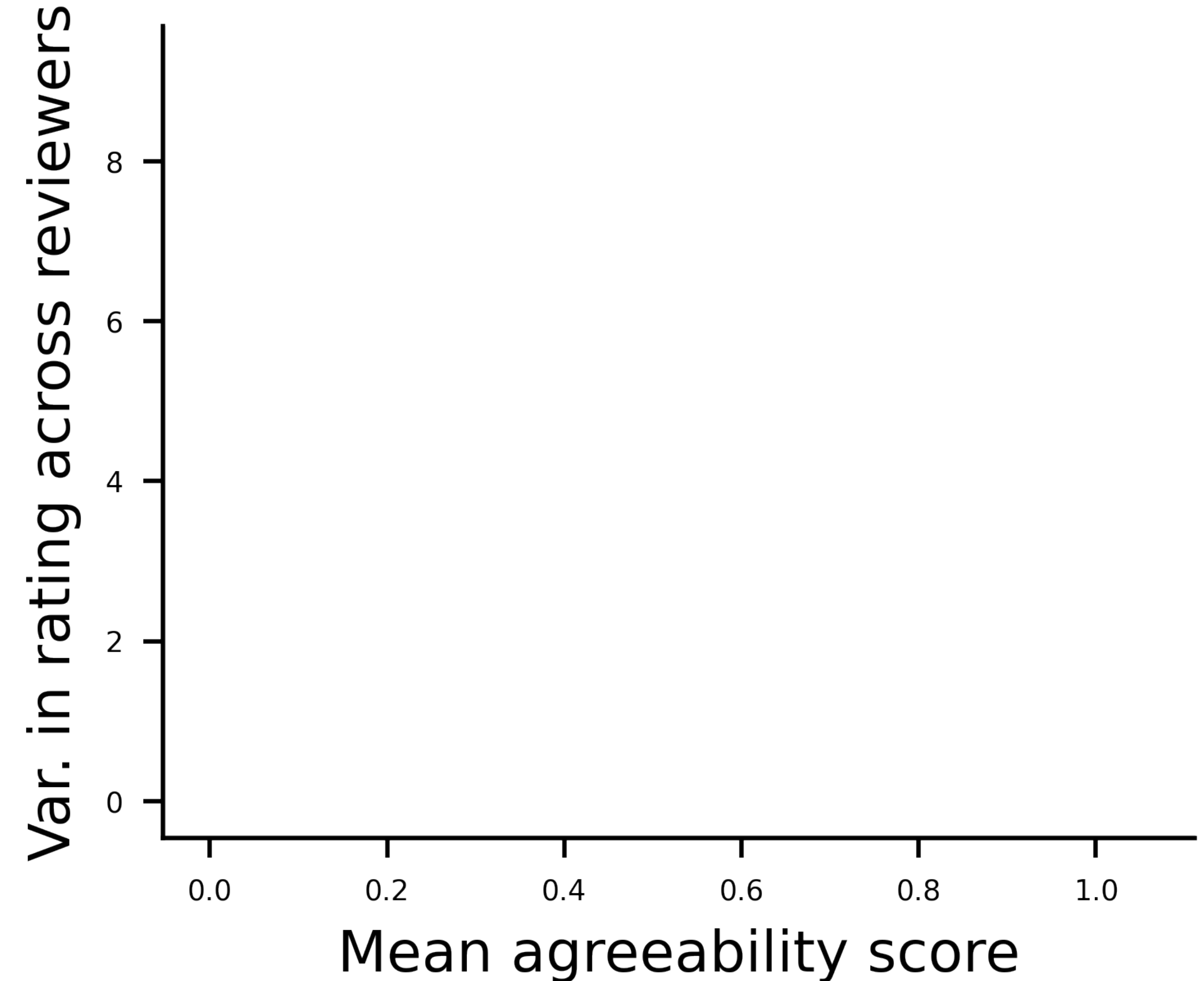
Out of argumentative statements, % of time author accepts premise

$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$

Agreeability

Agreeability reveals controversies not apparent from score variance.

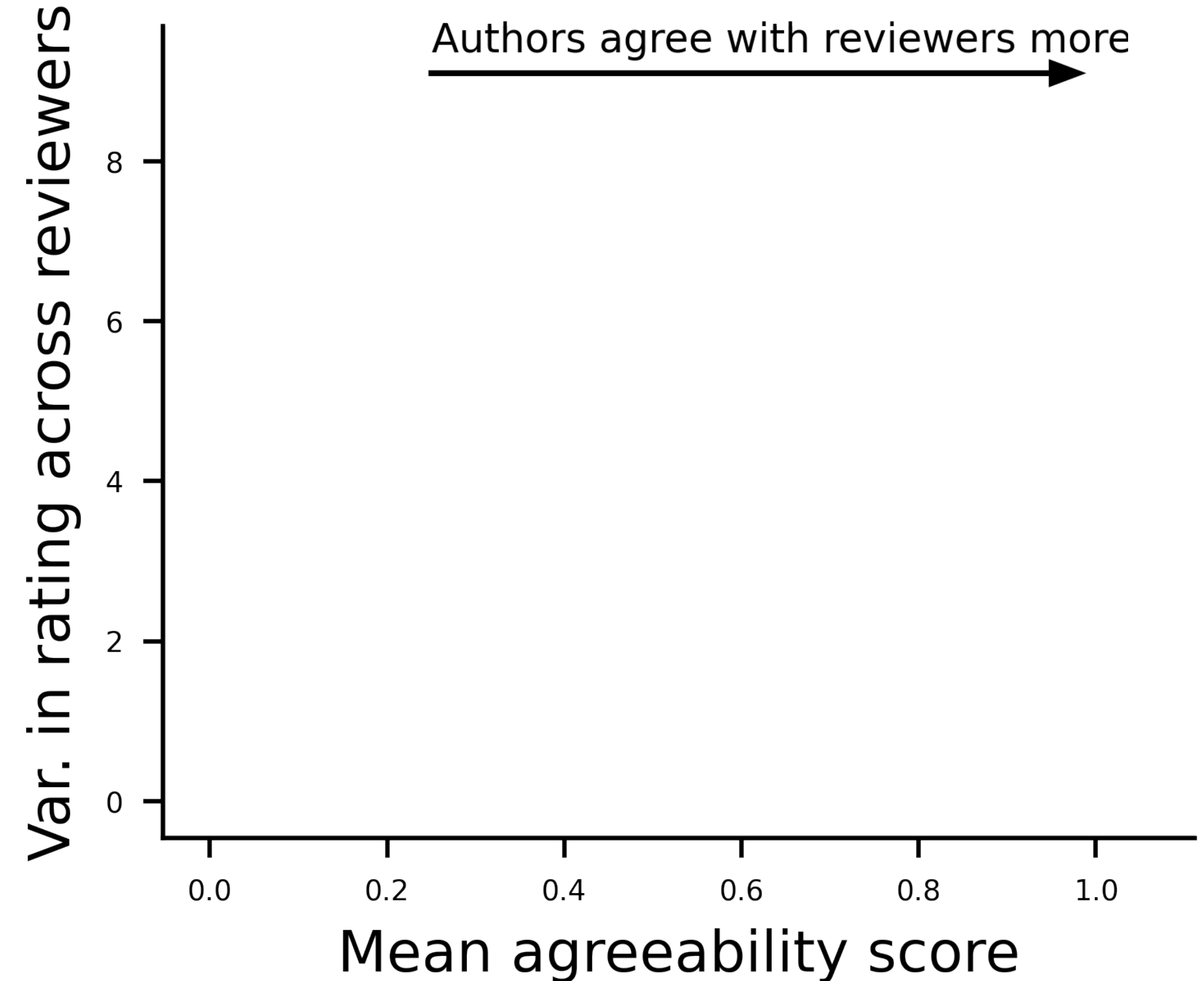
$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$



Agreeability

Agreeability reveals controversies not apparent from score variance.

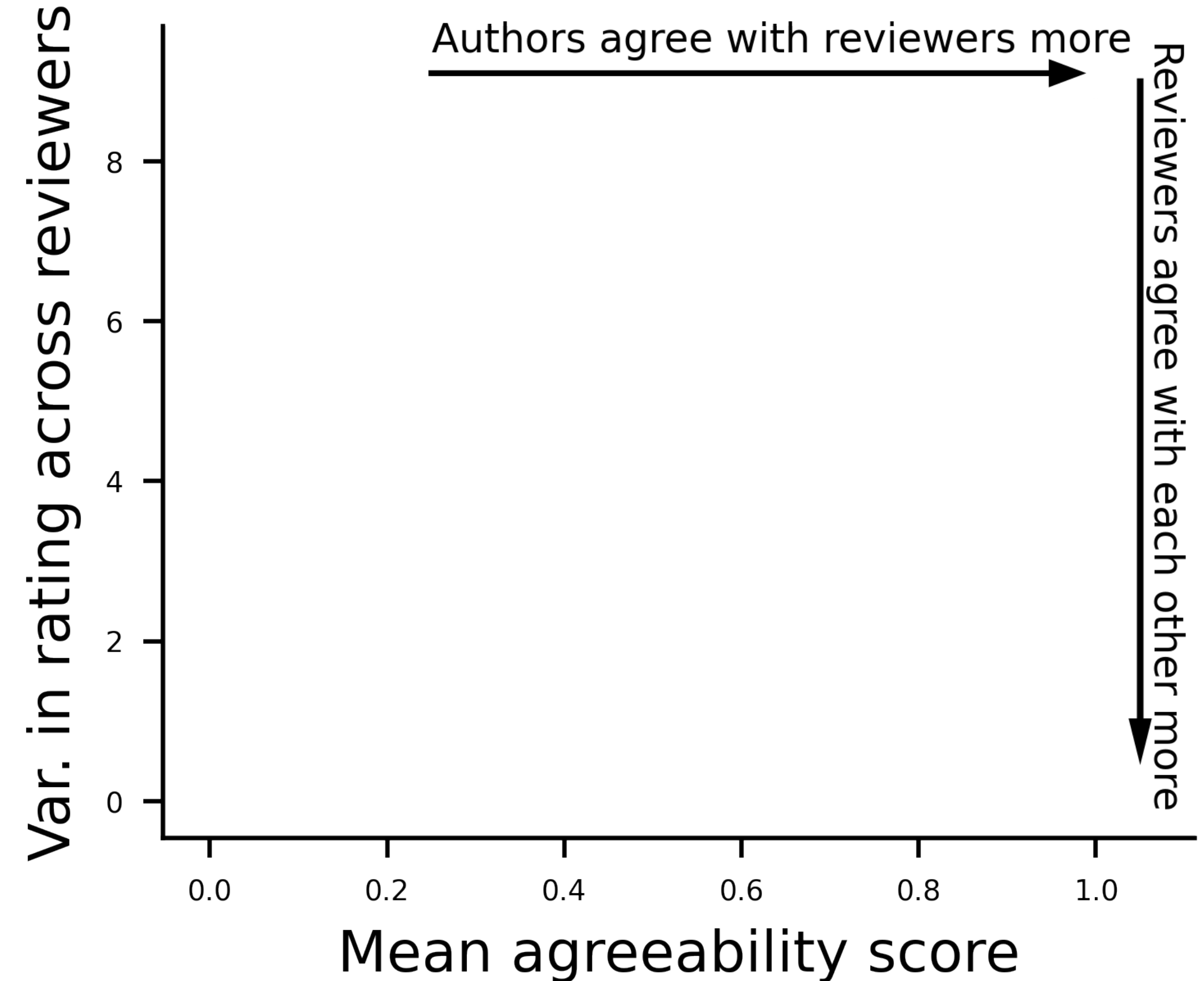
$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$



Agreeability

Agreeability reveals controversies not apparent from score variance.

$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$



Agreeability

Agreeability reveals controversies not apparent from score variance.

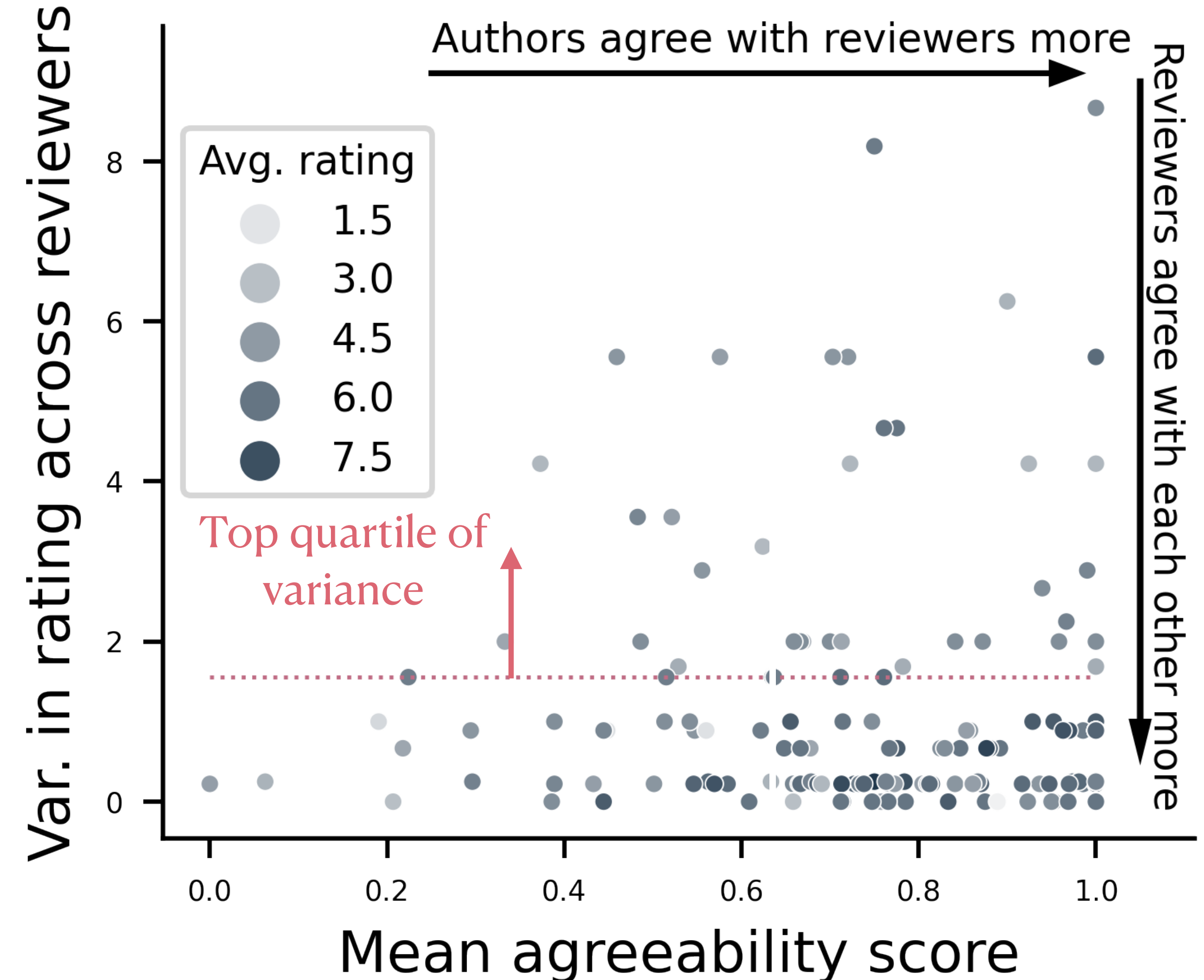
$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$



Agreeability

Agreeability reveals controversies not apparent from score variance.

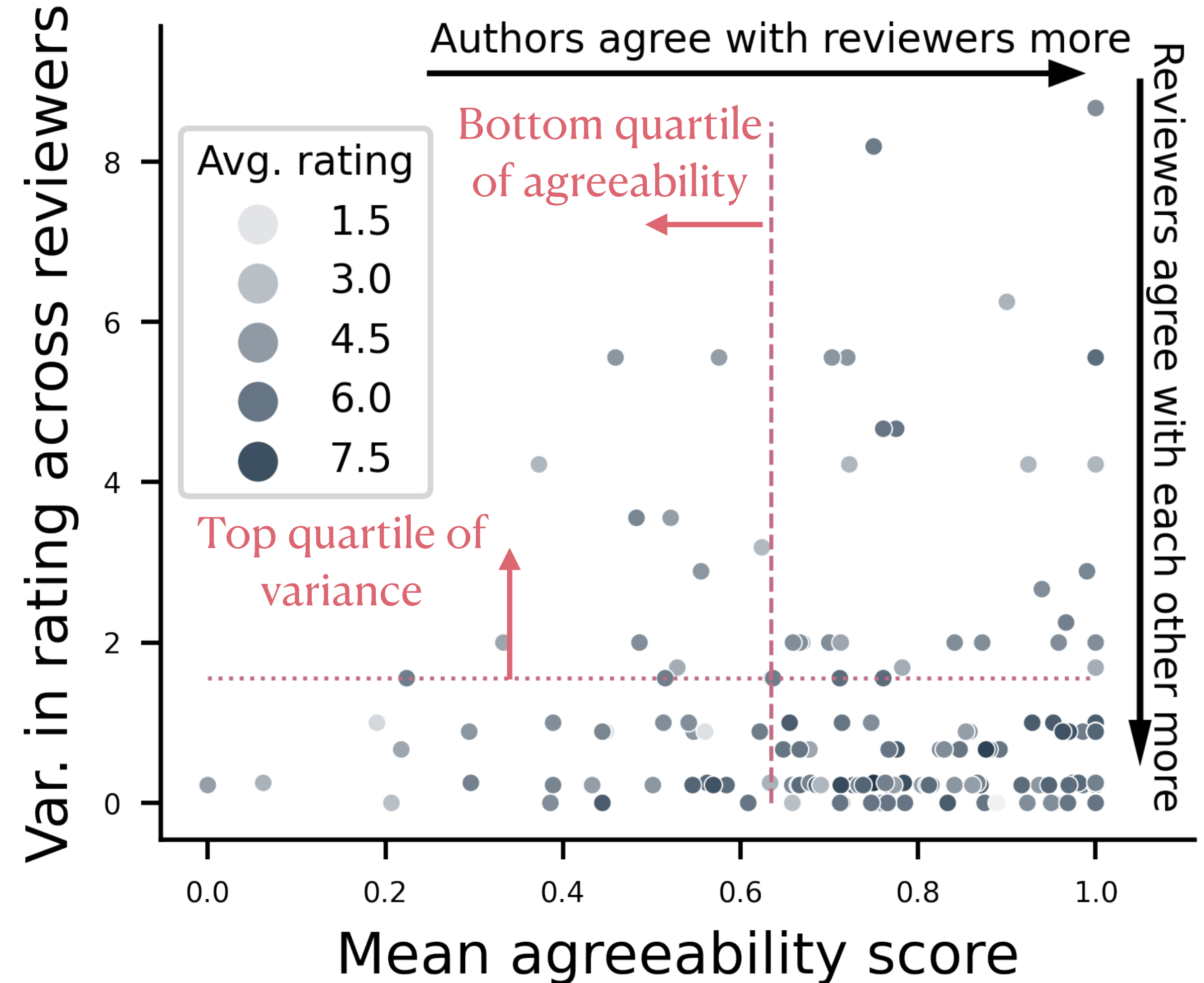
$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$



Agreeability

Agreeability reveals controversies not apparent from score variance.

$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$

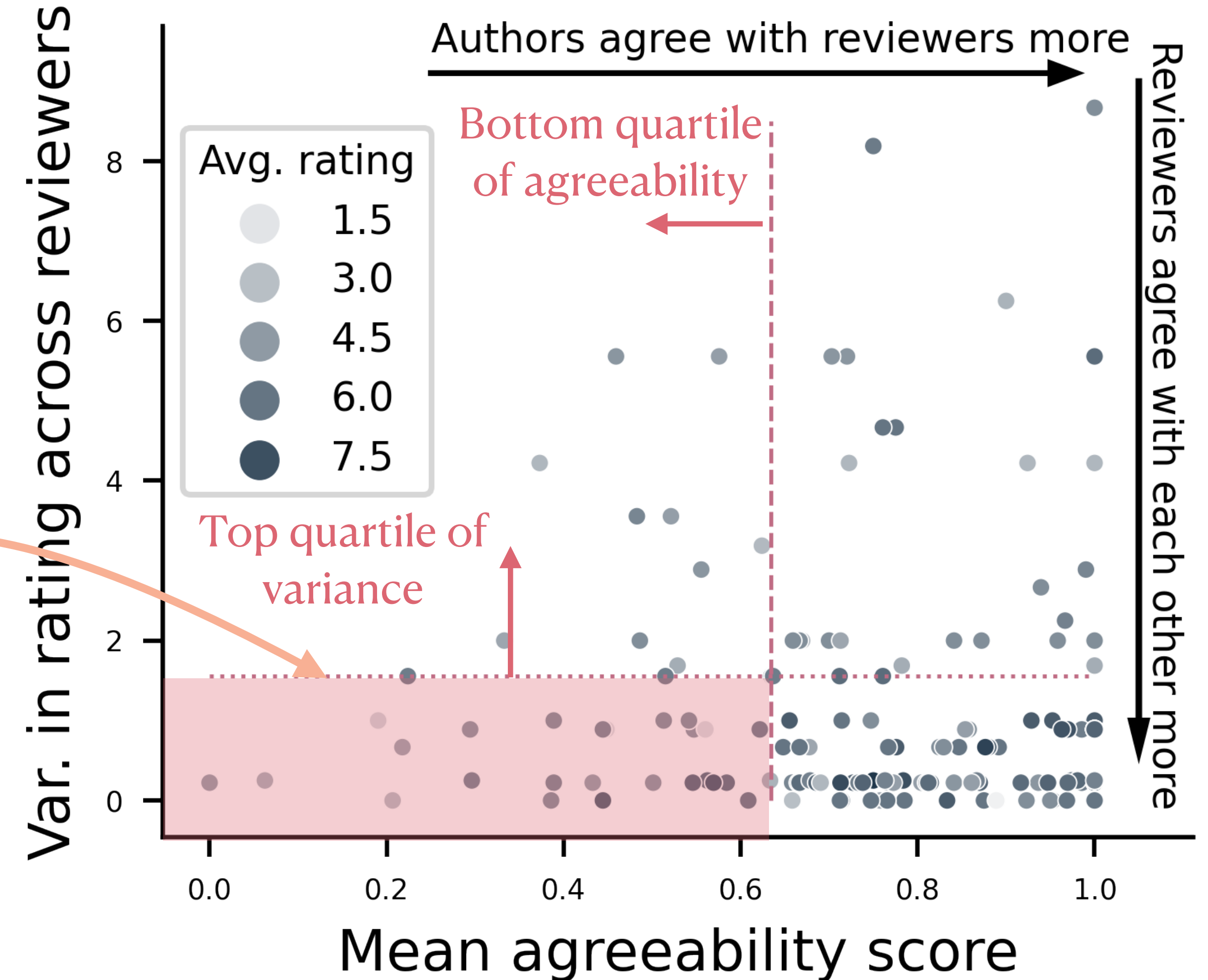


Agreeability

Agreeability reveals controversies not apparent from score variance.

$$agreeability = \frac{n_{concur}}{n_{concur} + n_{dispute}}$$

28/159
papers



DISAPERE

A dataset revealing nuances of peer review discussions

DISAPERE

A dataset revealing nuances of peer review discussions

How do we use these labels?

DISAPERE

A dataset revealing nuances of peer review discussions

How do we use these labels?

Post-hoc analysis

Designing policies and interfaces

DISAPERE

A dataset revealing nuances of peer review discussions

How do we use these labels?

- Post-hoc analysis

- Designing policies and interfaces

What about other (non-ML) domains?

DISAPERE

A dataset revealing nuances of peer review discussions

How do we use these labels?

Post-hoc analysis

Designing policies and interfaces

What about other (non-ML) domains?

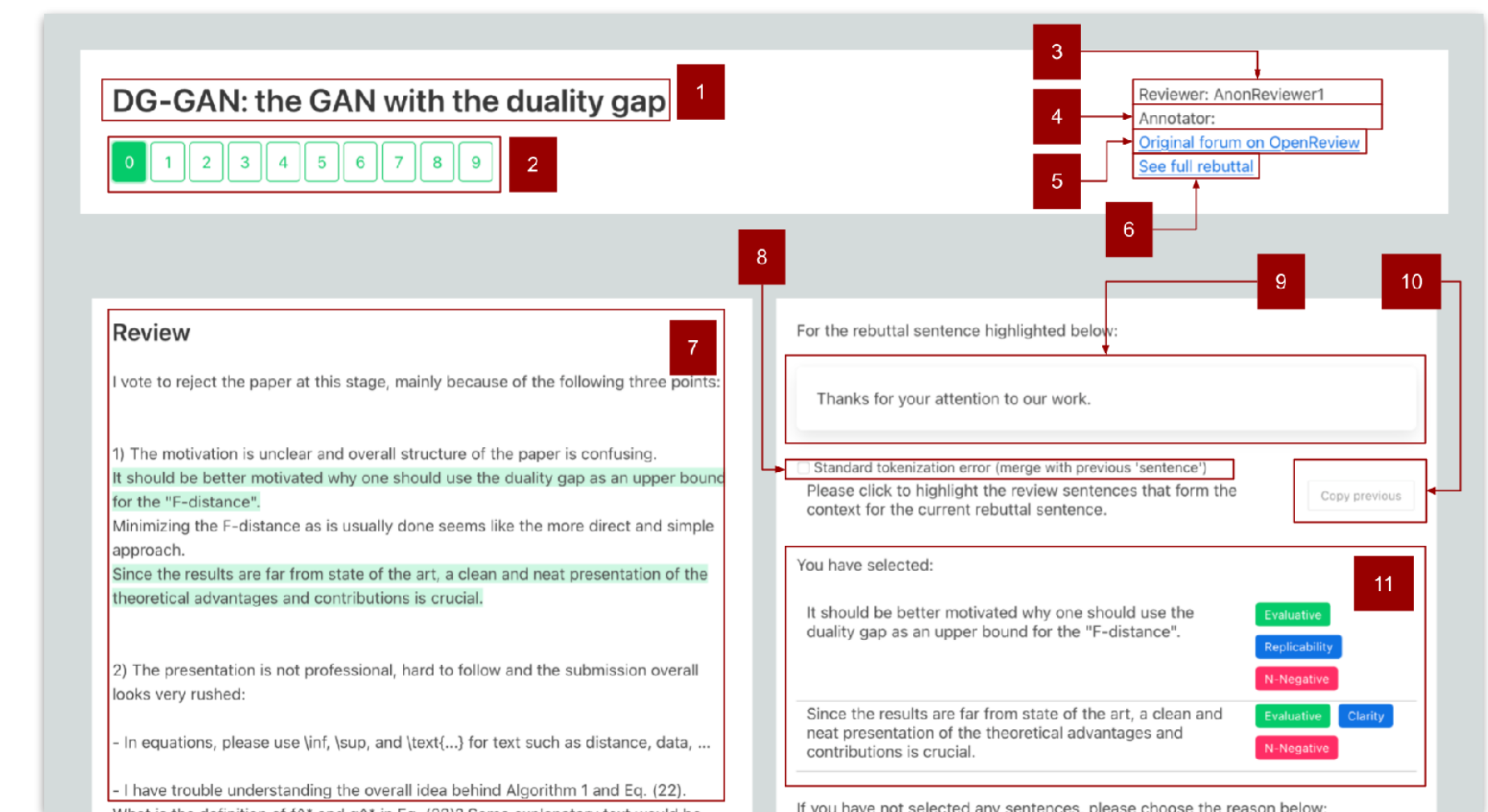
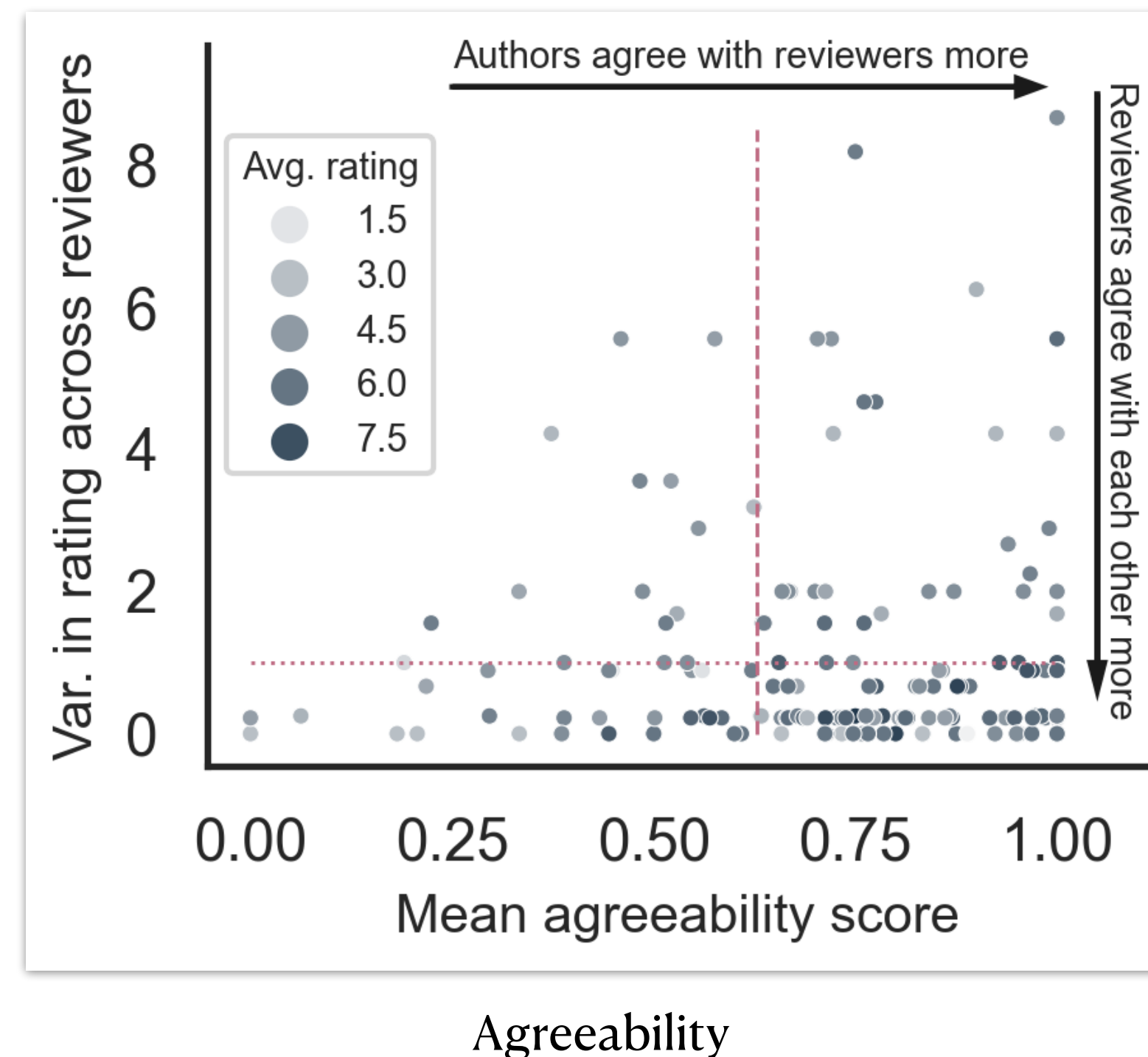
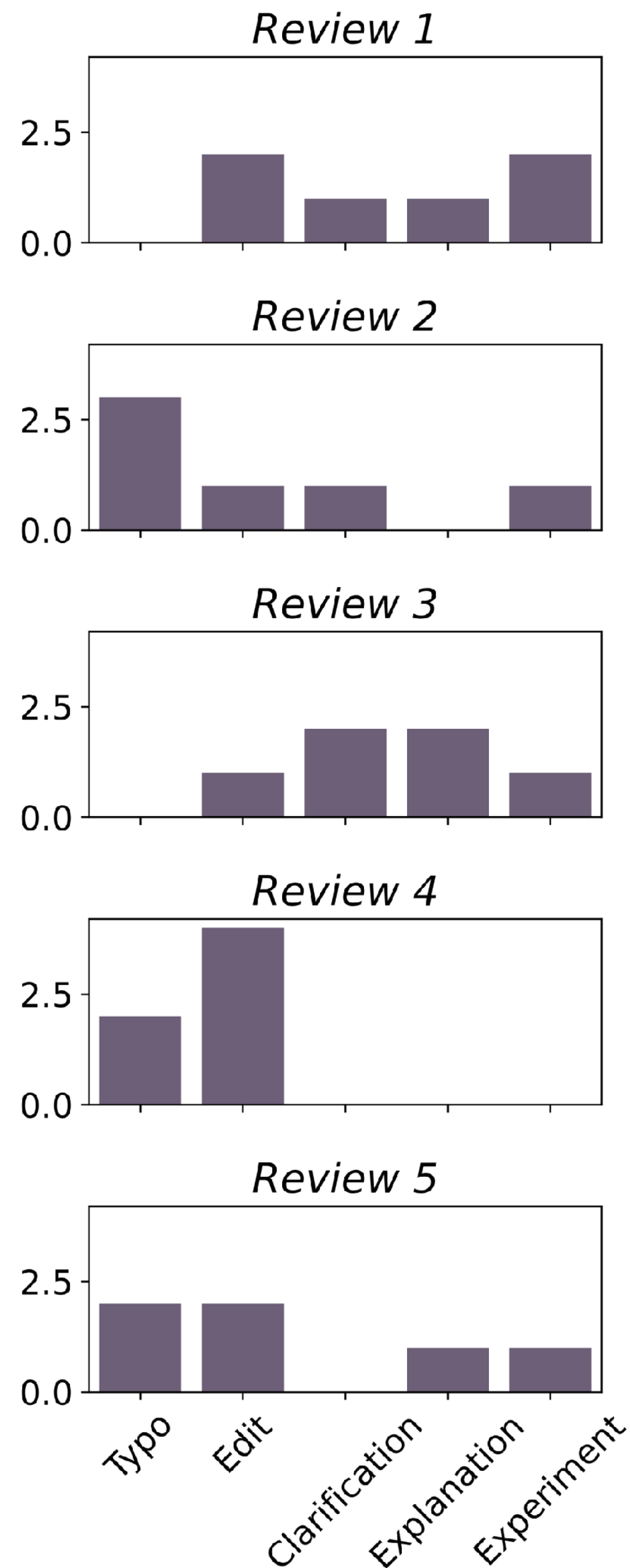
www.github.com/nnkennard/DISAPERE

The screenshot displays a web interface for the DISAPERE dataset. At the top, a title bar reads "DG-GAN: the GAN with the duality gap" (labeled 1). Below it is a sequence of green boxes numbered 0 through 9, with a red box labeled 2 to the right. On the right side, a sidebar contains a "Reviewer: AnonReviewer1" field, an "Annotator:" field, and links for "Original forum on OpenReview" and "See full rebuttal".

The main content area is divided into two sections. The left section, labeled 7, is titled "Review" and contains the text: "I vote to reject the paper at this stage, mainly because of the following three points:". It lists three points, with the first point highlighted in green and labeled 8. The first point discusses the motivation and structure of the paper, mentioning the "F-distance" and the "duality gap". The second point discusses the presentation of the paper, and the third point discusses the overall idea behind Algorithm 1 and Eq. (22). The right section, labeled 9, is titled "Rebuttal" and contains the text: "Thanks for your attention to our work." Below this, a sentence is highlighted in green and labeled 10: "Standard tokenization error (merge with previous 'sentence')". To the right of this sentence is a "Copy previous" button. Below the rebuttal, a section labeled 11 shows a list of selected sentences with corresponding evaluation buttons: "Evaluate", "Replicability", "N-Negative", "Clarity", and "N-Negative".

Thank you!

Tim O’Gorman, Rajarshi Das, Akshay Sharma , Chhandak Bagchi
Matthew Clinton , Pranay Kumar Yelugam, Hamed Zamani, Andrew McCallum



Software

www.github.com/nnkennard/DISAPERERE

kennard@cs.umass.edu



@nnkennard

Was the feedback **constructive**?

Was the feedback **applicable**?